

Dynamic global-local shrinkage process priors for time-varying parameter models

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Abstract

Introduction This thesis explores the application of a dynamic shrinkage process in a time-varying parameter model setting, using a probabilistic programming language - Stan. Understanding the feasibility of implementing such a model in Stan is very important, since Stan models are easily extendable to more complex nonlinear and non-Gaussian models.

Research Question The main research question is: “Is it possible to implement a time-varying parameter model with a dynamic shrinkage process in a probabilistic programming language like Stan to efficiently sample from the posterior using Hamiltonian Monte Carlo (HMC); and how would it perform compared to the established Gibbs sampling in terms of sampling efficiency, computing time, and accuracy?”

Method This study was divided in two phases: firstly, both Stan/HMC and the Gibbs sampling approach were tested and compared using a synthetic dataset to guarantee a controlled environment; secondly, the algorithms were also compared using a real-world dataset to understand the practicality of both.

Results The results indicate that Stan is a viable alternative to Gibbs sampling; as Stan consistently produces smoother and less variable posterior distributions. Nevertheless, the Gibbs sampling exhibits a shorter computation time and has better sampling efficiency.

Discussion These findings suggest that it is feasible to implement a time-varying parameter model with a dynamic shrinkage process in Stan. The preference between both implementations depends on the priorities of the user; if the priority is computation time, the most logical implementation to use is Gibbs. On the other hand, if the priority is the behaviour of the posterior distribution, the most likely option is Stan.

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Abbreviations

Abbreviation	Description
LGSSM	Linear Gaussian state-space model
FFBS	Forward filtering backward sampling
TVP	Time-varying parameters
TVR	Time-varying parameter regression
AR	Autoregressive model
CDF	Cumulative distribution function
PG	Pólya-Gamma distribution
MCMC	Markov chain Monte Carlo
MH	Metropolis-Hastings
HMC	Hamiltonian Monte Carlo
RW-MH	Random Walk Metropolis-Hastings
HPD	Highest posterior density
N_{eff}	Effective Sample Size
MCSE	Monte Carlo standard error
IID	Independent and identically distributed
SV	Stochastic volatility
CAPM	Capital Asset Pricing Model

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1

Introduction

The main objective when creating a model is to be able to capture the patterns present in the data. Therefore, the model must be flexible enough to capture different patterns. Nevertheless, if the model is allowed to be very flexible, overfitting may occur. When overfitting occurs, the model loses the ability to generalize to unseen data, i.e., the model has memorized the data rather than learning the general patterns in the data. Hence, the inference capabilities of the model on unseen data will decrease.

Regularization is one of many methods available in the current literature to prevent overfitting. There are different regularization methods, nevertheless all methods are built towards the same goal: control model complexity, and, consequently, improve the ability to generalize. The main difference between these methods is the way they introduce a penalty to control the model complexity. For instance, one of the earliest regularization methods in regression models is Ridge regression, which uses a quadratic penalty. In this method, the regression coefficients are shrunk toward zero to avoid overfitting. The shrinkage is controlled by a single global shrinkage parameter, which can lead to overshrinking of the true signals and under-shrinking of the noise.

In a Bayesian context, the regularization can be applied through the choices of the priors. In the case of Ridge regression, the prior for the regression coefficients is a Gaussian prior with the same variance for all regression coefficients. Ridge regression is a global regularization process. However, there are other types of regularization, such as global-local regularization; this process introduces two sets of parameters, one global for

all coefficients and one local for each individual coefficient, allowing the usual global regularization while incorporating a specific regularization for each coefficient. This approach reduces the risk of overshrinking the true signals and undershrinking the noise.

The global-local shrinkage priors are a well-established process that induces shrinkage and regularization in various types of models and applications; they were originally developed for implementation in static regression models. In general, these priors are constructed as continuous scale mixtures of Gaussian distributions; this type of construction yields desirable shrinkage properties, such as smoothness and sparsity (approximately).

Recently, these priors have been extended to time series regression models with time-varying parameters. Time-varying parameter models use a prior process to model the evolution of parameters over time. In this new dynamic setting, a parameter may remain constant over long periods of time, or it may change rapidly or even jump. This framework allows for greater flexibility in the parameters, if necessary.

In the current literature, there are several approaches of dynamic shrinkage priors, including the dynamic normal-gamma prior (Kalli and Griffin [12]), the dynamic shrinkage process prior (Kowal, Matteson, and Ruppert [15]), and the dynamic triple gamma prior (Knaus and Frühwirth-Schnatter [14]). The focus of the present study will be the dynamic shrinkage process prior proposed by Kowal, Matteson, and Ruppert [15]; the extension to the other two priors mentioned above is straightforward and is left for future studies.

The dynamic shrinkage process defines a framework for modelling time-varying parameters using a dynamic global-local shrinkage process as a prior. The shrinkage process is based on two components, the global shrinkage parameter and the set of local shrinkage parameters. The global parameter controls the overall shrinkage, while the local parameters control the shrinkage at each time-step. This structure allows the parameters to evolve over time with flexible dynamics.

In terms of computation, the approach proposed by Kowal, Matteson, and Ruppert [15] is based on a Gibbs sampling algorithm. This algorithm samples from the posterior distribution of the time-varying and static parameters; due to autocorrelation in the draws, a large number of samples is typically needed to reach convergence. This drawback raises the research question: “Is it possible to implement a time-varying parameter model with a dynamic shrinkage process in a probabilistic programming language

like Stan to efficiently sample from the posterior using Hamiltonian Monte Carlo (HMC); and how would it perform compared to the established Gibbs sampling in terms of sampling efficiency, computing time, and accuracy?”. HMC may be a viable alternative since it is known to sample efficiently in high-dimensional parameter spaces. Importantly, a workable Stan implementation for the Gaussian time-varying regression can be easily extended to non-Gaussian models; these extensions may be hard or even impossible for Gibbs sampling. HMC is known to be numerically expensive, particularly for models with challenging geometry like time-varying parameter models with global-local priors. It is therefore of interest to compare the advantages and disadvantages of Gibbs sampling and HMC in Stan in a controlled setting.

The main objective of this study is to determine whether the Stan approach is a reliable and efficient option, compared to the Gibbs sampling algorithm, for this class of models.

2

Extended Background

2.1 Time-varying parameter regression

The common approach when working with a time series regression model is to use parameters that are constant over time. Nevertheless, in some cases, this approach is not realistic and may be restrictive, as it implies that the data generating process remains unchanged over time. Therefore, this thesis will focus on regression models with time-varying parameters (TVP).

An example of a regression model with TVP is presented below, following the framework of Villani [25]:

$$y_t = \alpha_t + \beta_t x_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2) \quad (2.1)$$

$$\alpha_t = \alpha_{t-1} + \eta_t, \quad \eta_t \sim N(0, \sigma_\alpha^2) \quad (2.2)$$

$$\beta_t = \beta_{t-1} + \nu_t, \quad \nu_t \sim N(0, \sigma_\beta^2), \quad (2.3)$$

where y_t is a response variable, x_t is an explanatory variable, α_t is the intercept, β_t is the regression coefficient, and ϵ_t is the error term.

This model is characterized by the fact that both parameters, α_t and β_t , evolve according to a random walk process, see equations (2.2) and (2.3), where the main source of variation between time steps comes from the variances of the respective innovations, η_t and ν_t .

The regression model with TVP is conveniently analyzed in a state-space model framework, as explained in the coming sections.

2.2 Linear Gaussian state-space model

The linear Gaussian state-space model is the most widely used and studied state-space model. This model has a special property that allows all relevant posterior and predictive distributions to be expressed in closed form; exact estimation is therefore possible without resorting to approximation techniques.

The linear Gaussian state-space model (LGSSM) can be defined as:

$$\text{Measurement model: } y_t = Cz_t + \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_\epsilon) \quad (2.4)$$

$$\text{State dynamics: } z_t = Az_{t-1} + \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_\eta), \quad (2.5)$$

where y_t is a m -dimensional observation, or measurement, vector, z_t is a s -dimensional state vector, A is a $s \times s$ matrix that determines the transition of the state vector over time, C is a $m \times s$ matrix that maps the state vector to the observation vector, ϵ_t is the measurement error and η_t is the state innovation. The treatment in this thesis uses the notation for state-space models in Villani [24].

There are many nonlinear and non-Gaussian state space models. For example, a nonlinear state-space model with Gaussian measurement errors and innovations is of the form

$$\text{Measurement model: } y_t = C(z_t) + \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_\epsilon) \quad (2.6)$$

$$\text{State dynamics: } z_t = A(z_{t-1}) + \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_\eta), \quad (2.7)$$

where $C(\cdot)$ and $A(\cdot)$ are now nonlinear functions of the state vector. All models considered in this thesis can be recast into linear Gaussian models, so we will concentrate on this important subclass.

Model assumptions

The LGSSM makes four basic assumptions:

1. The state z_t is a first order Markov process; in other words, the current state z_t depends only on the previous state z_{t-1} .
2. The observations y_t are independent, given the current state vector z_t .
3. The measurement errors ϵ_t , the state innovations η_t , and the distribution for the initial state z_0 are all Gaussian (normal) distributions.

4. The state dynamics and the observation (measurement) model are both linear.

The initial state z_0 can be treated as known or assigned a prior distribution, for instance $z_0 \sim N(\mu_{0|0}, \Sigma_{0|0})$. This notation shows that the prior distribution is for the state at time $t = 0 \Rightarrow 0|0$, and is based on all that data up to time $t = 0 \Rightarrow 0|0$; this type of notation will be reused in the next subsections. The LGSSM can be further generalized, for example, by adding a control variable or exogenous variables, or allowing the model parameters to be time dependent.

2.2.1 Filtering in LGSSM

Filtering in state-space models refers to estimating the state z_t using data until time-step t . This thesis takes a Bayesian approach where filtering returns the *filtering posterior distribution*, $P(z_t|y_{1:t})$, where $y_{1:t} = (y_1, \dots, y_t)$ [24].

The filtering posterior in a general state-space model is computed with the Bayes filter. At each time-step, this update rule transforms the prior distribution $P(z_t|y_{1:t-1})$ into the posterior distribution $P(z_t|y_{1:t})$. This process is divided into two phases; see Villani [24]:

1. Prior propagation

$$P(z_t|y_{1:t-1}) = \int P(z_t|z_{t-1})P(z_{t-1}|y_{1:t-1})dz_{t-1} \quad (2.8)$$

2. Measurement update

$$P(z_t|y_{1:t}) \propto P(y_t|z_t)P(z_t|y_{1:t-1}) \quad (2.9)$$

Note that in the prior propagation step, the posterior distribution from the previous time $t - 1$ is used to compute the prior distribution at the current time t ; this process is called propagation, because it carries forward the previous posterior distribution.

The two steps of the Bayes filter update are usually infeasible, but the LGSSM is a tractable special case where it is possible to compute exact analytic expressions of the prior propagation and the measurement update; this property is the center of the famous *Kalman filter* described in the next subsection.

Kalman filter

The Kalman filter uses the same recursive logic of the traditional Bayes filter update, but instead of updating the full posterior distribution, it only updates the posterior mean and covariance matrix. This is only possible due to the fact that the filtering posterior in the LGSSM follows a Gaussian distribution, which is completely characterized by its mean and covariance; see Durbin and Koopman [5]:

$$z_t|y_{1:t} \sim N(\mu_{t|t}, \Omega_{t|t}) \quad (2.10)$$

This property holds, due to the assumptions of LGSSM; see Section 2.2.

Similarly to the Bayes filter update, the Kalman filter update is also divided in the same phases:

1. Prior propagation

$$N(\mu_{t-1,t-1}, \Omega_{t-1,t-1}) \Rightarrow N(\mu_{t,t-1}, \Omega_{t,t-1})$$

$$\mu_{t,t-1} = A\mu_{t-1,t-1}; \quad \Omega_{t,t-1} = A\Omega_{t-1,t-1}A^\top + \Sigma_\eta \quad (2.11)$$

2. Measurement update

$$N(\mu_{t,t-1}, \Omega_{t,t-1}) \Rightarrow N(\mu_{t,t}, \Omega_{t,t})$$

$$\mu_{t,t} = \mu_{t,t-1} + K_t(y_t - C\mu_{t,t-1}); \quad \Omega_{t,t} = (I - K_tC)\Omega_{t,t-1} \quad (2.12)$$

The Kalman gain K_t specifies how much importance the most recent measurement y_t , or rather the most recent prediction error $y_t - C\mu_{t,t-1}$, has compared to the prior $\mu_{t,t-1}$.

$$K_t = \Omega_{t,t-1}C^\top(C\Omega_{t,t-1}C^\top + \Sigma_\epsilon)^{-1} \quad (2.13)$$

2.2.2 Smoothing in LGSSM

Smoothing is another inference method used in state-space models. Smoothing produces an estimate $\hat{z}_{t|T}$, using *all* data until the last time step T . In a Bayesian context, the smoothing objective is to compute the *smoothing posterior distribution*, $P(z_t|y_{1:T})$; this is a retrospective posterior that infers the state at a distant time in the past. The retrospective posterior is also known as *marginal* smoothing posterior since it refers to a single time point t ; the joint smoothing posterior is defined as $P(z_{1:T}|y_{1:T})$ [24].

To sample from the joint smoothing posterior, it is common to use the forward filtering and backward sampling (FFBS) algorithm [3, 8]. FFBS is

an algorithm that also uses the Kalman filter, in this setting it is used to compute all the means $\mu_{t|t}$ and variances $\Omega_{t|t}$ for each time-step t for $t = 1, \dots, T$ (forward filtering). The filtering distribution for the last state $z_T \sim N(\mu_{T|T}, \Omega_{T|T})$ is then used as a starting point for the backward sampling: $z_t|z_{t+1} \sim N(\mu_{t|t+1}, \Omega_{t|t+1})$ for $T, T-1, \dots, 1$. The output of the FFBS algorithm will be draws $z_{1:T}$ from the joint smoothing posterior $P(z_{1:T}|y_{1:T})$. Furthermore, this process is scalable, since instead of simulating only one final state z_T it is possible to simulate N final states, and consequently obtain N independent draws of the whole trajectory $z_{1:T}$.

The complete FFBS algorithm is presented for the sake of completeness and clarity, following the framework presented in Villani [24].

The FFBS algorithm

0. Input:

- (a) Time series: $y_{1:T} = (y_1, \dots, y_T)$
- (b) Model parameters $A, C, \Sigma_\eta, \Sigma_\epsilon$
- (c) Initial state prior mean and covariance matrix ($\mu_{0|0}$ and $\Omega_{0|0}$)

1. Apply the Kalman Filter using the time series, model parameters, and the initial state prior mean and covariance. This will produce $(\mu_{t|t}, \Omega_{t|t}, \mu_{t|t-1}, \Omega_{t|t-1})$ for all $t = 1, \dots, T$

2. Simulate z_T from the final filtering distribution $z_T \sim N(\mu_{T|T}, \Omega_{T|T})$

3. Simulate backward in time for $t = T-1, T-2, \dots, 1$:

- (a) $\mu_{t|t+1} = \mu_{t|t} + \Omega_{t|t} A^\top \Omega_{t+1|t}^{-1} (z_{t+1} - \mu_{t+1|t})$
- (b) $\Omega_{t|t+1} = \Omega_{t|t} - \Omega_{t|t} A^\top \Omega_{t+1|t}^{-1} A \Omega_{t|t}$
- (c) Simulate $z_t|z_{t+1} \sim N(\mu_{t|t+1}, \Omega_{t|t+1})$

4. Return draw of $z_{1:T}$ from the joint smoothing posterior $P(z_{1:T}|y_{1:T})$ conditional on the model parameters.

If the number of simulations of the final state is different than one, the algorithm will repeat steps 2 and 3 to produce multiple draws $z_{1:T}^{(1)}, \dots, z_{1:T}^{(N)}$.

2.2.3 Parameter inference in LGSSM

Until this section, it has been assumed that the model parameters $A, C, \Sigma_\epsilon, \Sigma_\eta$ are known. There are many applications where this assumption does not

hold, and it is then necessary to estimate the model parameters from the data. Let the model parameters be collected in a vector θ . There are two main approaches for sampling from the joint posterior distribution $P(\theta, z_{1:T}|y_{1:T})$ of the model parameters θ and the states $z_{1:T}$ (Villani [24]). The first approach decomposes the joint posterior distribution using a marginal-conditional factorization:

$$P(\theta, z_{1:T}|y_{1:T}) = P(z_{1:T}|\theta, y_{1:T})P(\theta|y_{1:T}), \quad (2.14)$$

Samples from the marginal posterior distribution $P(\theta|y_{1:T})$ can be obtained with, for example, the Metropolis-Hastings algorithm. The state trajectory $z_{1:T}$ can then be sampled using the FFBS algorithm and the simulated draws $\theta^{(j)}$. The second approach uses a two-block Gibbs sampler, sampling θ and $z_{1:T}$ from the conditionals distributions, $P(\theta|z_{1:T}, y_{1:T})$ and $P(z_{1:T}|\theta, y_{1:T})$, respectively.

Sampling using the marginal-conditional factorization

The first approach will now be presented in more detail. Given Bayes' Theorem, the marginal posterior can be written as:

$$P(\theta|y_{1:T}) \propto P(y_{1:T}|\theta)P(\theta), \quad (2.15)$$

where $P(\theta)$ is the prior distribution for θ . Villani [24] shows that the marginal likelihood can be written as:

$$P(y_{1:T}|\theta) = \prod_{t=1}^T N(y_t|C\mu_{t|t-1}, C\Omega_{t|t-1}C^\top + \Sigma_\epsilon) \quad (2.16)$$

Note that $\mu_{t|t-1}$ and $\Omega_{t|t-1}$ are computed in the prior propagation step of the Kalman filter; see Section 2.2.1. Therefore, the marginal likelihood can be evaluated using the Kalman filter at each time-step ($t = 1, \dots, T$). Following the framework described in Villani [24], here is the algorithm for evaluating the marginal likelihood (in logarithmic scale for numerical stability).

The marginal likelihood algorithm based on the Kalman filter

0. Input:

- (a) Time series: $y_{1:T} = (y_1, \dots, y_T)$
- (b) Model parameters $\theta = (A, C, \Sigma_\eta, \Sigma_\epsilon)$

- (c) Initial state prior mean and covariance matrix ($\mu_{0|0}$ and $\Omega_{0|0}$)
- 1. Apply the Kalman Filter using the time series, model parameters, and the initial state prior mean and covariance. This will produce $(\mu_{t|t-1}, \Omega_{t|t-1})$ for all $t = 1, \dots, T$
- 2. Set $\ell = 0$. Iterate for t in $1, \dots, T$:
 - (a) $\ell = \ell + \log N(y_t | C\mu_{t|t-1}, C\Omega_{t|t-1}C^\top + \Sigma_\epsilon)$
- 3. Return: Log marginal likelihood $\log(P(y_{1:T}|\theta))$

Now, all the conditions to use the Metropolis-Hastings algorithm are met, and it is possible to generate draws of the model parameters $\theta^{(j)}$. Note that without the marginal likelihood algorithm, it would not be possible to use the Metropolis-Hastings algorithm since the marginal likelihood needs to be computable.

The Metropolis-Hastings algorithm proposes draws from a proposal distribution and then accepts or rejects that draw based on the acceptance probability α .

The Metropolis-Hastings algorithm

- 0. Input:
 - (a) Time series: $y_{1:T} = (y_1, \dots, y_T)$
 - (b) Number of posterior draws: N
 - (c) Number of burn-in draws: b
 - (d) Proposal distribution $Q(\theta^*|\theta^{(k-1)})$
 - (e) Prior: $P(\theta)$
 - (f) Initial value: $\theta^{(0)}$ and $\log(P(y|\theta^{(0)}))$
- 1. for k in $1, \dots, (m + b)$:
 - (a) Simulate a proposal draw $\theta^* \sim Q(\theta^*|\theta^{(k-1)})$
 - (b) Compute $r = \log(P(y|\theta^*) - \log(P(y|\theta^{(k-1)}) + \log(P(\theta^*)) - \log(P(\theta^{(k-1)}))) + \log(Q(\theta^{(k-1)}|\theta^*)) - \log(Q(\theta^*|\theta^{(k-1)})))$ and $\alpha = \min(0, e^r)$
 - (c) Simulate $u \sim \text{Uniform}(0, 1)$
 - i. if $u < \alpha$ set $\theta^{(k)} = \theta^*$
 - ii. if $u \geq \alpha$ set $\theta^{(k)} = \theta^{(k-1)}$

2. Return: draws of the model parameters $\theta^{(j)}$ from the marginal posterior $P(\theta|y_{1:T})$, $\theta^{(j)} = \theta^{(b+1)} + \dots + \theta^{(b+m)}$.

The proposal distribution was evaluated on the logarithmic scale because the marginal likelihood was also evaluated on that scale.

Since samples of the model parameters $\theta^{(j)}$ are now available, it is possible to sample the states $z_{1:T}$ as well. As defined in equation (2.14), the marginal posterior for the states $z_{1:T}$ is: $P(z_{1:T}|\theta, y_{1:T})$. This conditional posterior is equivalent to the joint smoothing posterior, already seen in Section 2.2.2, but with the true model parameters as unknowns. Therefore, it is possible to sample the states $z_{1:T}$ using the FFBS algorithm, but now with each state trajectory draw conditioned on different simulated parameter draws $\theta^{(1)}, \dots, \theta^{(N)}$. The final result will be N samples from the joint posterior (2.14) of the states and model parameters $(\theta^{(j)}, z_{1:T}^{(j)})$.

Gibbs sampling

This section describes the two-block Gibbs sampler. In the first block, the model parameters $\theta^{(j)}$ will be updated using the old values of the states $z_{1:T}^{(j-1)}$.

$$\theta^{(j)}|z_{1:T}^{(j-1)} \sim P(\theta|z_{1:T}^{(j-1)}, y_{1:T})$$

In the second block the values of the states $z_{1:T}^{(j)}$ will be updated based on the newly updated model parameters $\theta^{(j)}$

$$z_{1:T}^{(j)}|\theta^{(j)} \sim P(z_{1:T}|\theta^{(j)}, y_{1:T})$$

The complete two-block Gibbs sampler is presented for the sake of completeness and clarity.

The two-block Gibbs algorithm

0. Input:

- (a) Initial values: $z_{1:T}^{(0)}$
- (b) Time series: $y_{1:T} = (y_1, \dots, y_T)$
- (c) Number of posterior draws - m

1. for k in $1, \dots, m$:

- (a) $\theta^{(k)} \sim P(\theta^{(k)}|z_{1:T}^{(k-1)}, y_{1:T})$
- (b) $z_{1:T}^{(k)} \sim P(z_{1:T}^{(k)}|\theta^{(k)}, y_{1:T})$

2. Return m correlated samples of $(\theta^{(k)}, z_{1:T}^{(k)})$ that converge in distribution to $P(\theta, z_{1:T}|y_{1:T})$.

The draws from $z_{1:T}^{(k)}$ in step 1.(b) of the Gibbs algorithm can be done by the FFBS algorithm.

There are noteworthy advantages and drawbacks to this method. On one hand, the posterior distribution used in step 1.(a) of the Gibbs algorithm can generally be computed in closed form using conjugate priors, without resorting to a time-consuming Kalman filter. On the other hand, the Gibbs sampling is typically slow to converge if the states $z_{1:T}$ and model parameters θ are highly correlated in the posterior, leading to a large number of draws to reach convergence.

2.2.4 Time-varying parameter regression in LGSSM

As stated in Section 2.2, the LGSSM framework allows for the model parameters to be time-varying; note that until now, only the state z_t , measurement error ϵ_t , and state innovation η_t were allowed to be time-varying. Therefore, the LGSSM can also be presented as follows:

$$\begin{aligned} \text{Measurement model: } y_t &= C_t z_t + \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_{t,\epsilon}) \\ \text{State dynamics: } z_t &= A_t z_{t-1} + \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_{t,\eta}). \end{aligned} \quad (2.17)$$

This new formulation opens different possibilities, one of which is time-varying parameter regression (TVR). To formulate a TVR it is necessary to re-parametrise the general LGSSM formulation (2.17); the changes necessary are $C_t = x_t^\top$, $A_t = I_p$, and $z_t = \beta_t$, assuming:

- x_t as a $1 \times (p + 1)$ vector $\Rightarrow x_t = (1, x_{1,t}, \dots, x_{p,t})$, p represents the number of parameters β_t of the regression.
- I_{p+1} as a $(p + 1) \times (p + 1)$ identity matrix.
- β_t as $(p + 1) \times 1$ vector $\Rightarrow \beta_t = (\beta_{0,t}, \dots, \beta_{p,t})$.

Thus, the TVR can be formulated as follows [24]:

$$y_t = x_t^\top \beta_t + \epsilon_t, \quad \epsilon_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_\epsilon) \quad (2.18)$$

$$\beta_t = \beta_{t-1} + \eta_t, \quad \eta_t \stackrel{\text{iid}}{\sim} N(0, \Sigma_\eta) \quad (2.19)$$

It is worth noting that:

- The variance of the errors Σ_ϵ and the variance of the innovations Σ_η are constant values over time.
- The matrix C_t is a deterministic function of time, since the covariates are always known.
- Since the TVR is a reparametrisation of the LGSSM, it benefits from the same properties.

The time-varying parameter regression formulated above; see equations (2.18) and (2.19), is the simplest approach for estimating the regression coefficients β_t , and consequently for modelling the response variable y_t . Nevertheless, there is a clear problem with this approach, specifically in the way the coefficients β_t evolve over time. Since the variance of the innovations Σ_η is constant over time, the sampled values of the innovations will evolve with the same constant scale, i.e., the innovations have the same variance for all t . To illustrate the parameter evolution, the trajectories of the parameter β_t were simulated with two different variances, following equation 2.19, as shown in Figure 2.1.¹

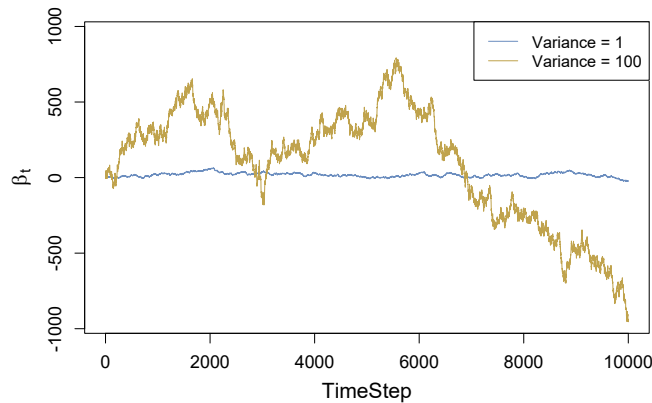


Figure 2.1: Simulated time-varying coefficient β_t

This is problematic because the model lacks the flexibility to capture heterogeneous behaviour of the coefficients β_t over time. For instance, the effect of an explanatory variable on the response may have three types of behaviour: constant positive impact in the early part of the sample, followed by a period of rapidly changing impact toward zero effect, and finally have zero impact in the latter part of the sample; we may also want to allow for sizeable jumps in the parameters. These types of behaviour cannot be

¹The R code required to reproduce Figure 2.1 is provided in a public GitHub repository

captured by the current model, because it is unable to adapt between periods of stable and abrupt changes. There have been several suggestions in the literature for resolving this lack of flexibility in the model; the present study focus on the dynamic shrinkage process prior (Kowal, Matteson, and Ruppert [15]) .

2.2.5 Dynamic shrinkage process prior

An interesting new prior for time-varying parameter models was recently proposed by Kowal, Matteson, and Ruppert [15] as a time series extension of the well-known horseshoe prior for regression [4]. This approach uses a more general class of dynamic shrinkage processes and has several special cases. Although the dynamic horseshoe process is the main focus given the scope of this thesis, the general approach will be described first, followed by the specification of the dynamic horseshoe process.

Consider the classical *local level model* with homoscedastic Gaussian innovations in the level of the time series

$$y_t = \beta_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2) \quad (2.20)$$

$$\beta_t = \beta_{t-1} + \omega_t, \quad \omega_t \sim N(0, \tau^2), \quad (2.21)$$

where the level β_t varies over time with Gaussian innovations ω_t and global homoscedastic innovation variance τ^2 . As expected, the behaviour of innovations ω_t has a direct influence on the path of β_t . When ω_t has a small variance, it is more likely to draw values close to zero, therefore $\beta_t \approx \beta_{t-1}$. On the other hand, when ω_t has a larger variance, it is more likely to draw larger values, so β_t will be more affected, $\beta_t \neq \beta_{t-1}$. As a result, the scale parameters play an important role in determining the magnitude of each innovation ω_t , and consequently, have an impact on the smoothness and adaptability of β_t . As explained above, the constant innovation variance can be limiting: the changes in the level β_t need to be roughly of the same size throughout time.

The dynamic shrinkage process generalizes the homoscedastic Gaussian innovation model in (2.21) to a global-local shrinkage prior of the form

$$\omega_t | \tau, \lambda_t \sim N(0, \tau^2 \lambda_t^2), \quad (2.22)$$

using a global variance τ^2 that is modified by local variance factors λ_t^2 at every time point. The global scale parameter τ controls the shrinkage that

is common to all innovations ω_t for all t , whereas the local scale parameters λ_t controls the shrinkage for each innovation ω_t . The choice of priors for the hyperparameters τ^2 and λ_t^2 should consider the dual requirement of accommodating large signals, while still being able to shrink the noise adequately. Usually, this property is defined as locally adaptive shrinkage.

The basis of the dynamic shrinkage process lies in allowing the local parameter λ_t to depend on the history of the shrinkage process; the framework proposed in Kowal, Matteson, and Ruppert [15] models the dependence between the local scale parameters λ_t on the logarithmic scale by defining $h_t = \log(\tau^2 \lambda_t^2)$ and the evolution of time as an autoregressive model (AR):

$$h_t = \mu + \phi(h_{t-1} - \mu) + \eta_t, \quad (2.23)$$

where ϕ is the autoregressive coefficient, and μ is the mean of the process. The next subsection presents a prior for the autoregressive coefficient, which ensures that $|\phi| < 1$, for the purposes of stationarity.

As shown in Kowal, Matteson, and Ruppert [15], if $\eta_t \stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1)$, the resulting dynamic shrinkage process will have similar sparsity properties as other priors in the literature. The $Z(\alpha, \beta, 0, 1)$ distribution is a heavy-tailed distribution defined as the distribution of the random variable $\log(X/(1-X))$ where $X \sim \text{Beta}(\alpha, \beta)$; more information on the Z distribution is available below. Now, if the hyper-parameters α and β are assigned the same value of $\frac{1}{2}$, the draws will have the same robustness and sparsity of the Horseshoe prior. Table 2.1 from [15] shows some special cases of the priors.

Parameters	Prior
$\alpha=\beta=\frac{1}{2}$	Horseshoe
$\alpha=\frac{1}{2}, \beta=1$	Strawderman-Berger
$\alpha=1, \beta=c-2, c>0$	Normal-exponential-gamma
$\alpha=\beta \rightarrow 0$	(Improper) normal-Jeffreys

Table 2.1: Special cases of global-local shrinkage priors

The local level model with the dynamic shrinkage process prior can be specified as follows:

$$\begin{aligned}
y_t &= \beta_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2) \\
\beta_t &= \beta_{t-1} + \omega_t, \quad \omega_t | h_t \sim N(0, \exp(h_t)) \\
h_t &= \log(\tau^2 \lambda_t^2) \\
h_t &= \mu + \phi(h_{t-1} - \mu) + \eta_t \\
\eta_t &\stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1).
\end{aligned} \tag{2.24}$$

Dynamic Shrinkage process for multivariate data

The dynamic shrinkage process prior can be applied to multiple regression with time-varying parameters [15]

$$\begin{aligned}
y_t &= x_t^\top \beta_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2), \quad \text{for } t = 1, \dots, T \\
\beta_t &= \beta_{t-1} + \omega_t, \quad \omega_{j,t} | h_{j,t} \stackrel{\text{indep}}{\sim} N(0, \exp(h_{j,t})) \\
h_{j,t} &= \log(\tau_0^2 \tau_j^2 \lambda_{j,t}^2), \quad \text{for } j = 1, \dots, p \\
h_t &= \mu + \Phi(h_{t-1} - \mu) + \eta_t, \quad \eta_{j,t} \stackrel{\text{iid}}{\sim} Z(\alpha, \beta, 0, 1) \\
\mu_j &= \log(\tau_0^2 \tau_j^2) \\
\sigma_\epsilon^2 &\propto \frac{1}{\sigma_\epsilon^2} \\
\tau_0 &\sim C^+\left(0, \frac{\sigma_\epsilon}{\sqrt{Tp}}\right), \quad \tau_j \stackrel{\text{indep}}{\sim} C^+(0, 1), \\
\frac{\phi_j + 1}{2} &\sim \text{Beta}(10, 2), \quad j = 1, \dots, p,
\end{aligned} \tag{2.25}$$

$$h_{j,0} \sim N(\mu_j, \xi_{j,0}^{-1}), \quad \xi_{j,0} \sim \text{PG}(1, 0),$$

where $\beta_t = (\beta_{1,t}, \dots, \beta_{p,t})$, $h_t = (h_{1,t}, \dots, h_{p,t})$, $\eta_t = (\eta_{1,t}, \dots, \eta_{p,t})$ and $\Phi = \text{diag}(\phi_1, \dots, \phi_p)$. It is worth noting that instead of the usual global parameter τ and local parameter λ_t , the multivariate approach extends the parameters: τ_0 indicates the global shrinkage time, and across all parameters, τ_j indicates the global shrinkage of the parameter j across time, and $\lambda_{j,t}$ indicates the local shrinkage associated with parameter j in time step t . It is also important to highlight that non-diagonal extensions are available for Φ , however for simplicity reasons this approach was chosen.

Dynamic horseshoe process

The dynamic horseshoe process was selected due to its locally adaptive shrinkage properties. In other words, it allows to shrink the noise, while maintaining the large signals. This builds on the characteristics of the static horseshoe prior, proposed in Carvalho, Polson, and Scott [4]. In addition to the properties of the static version, the shrinkage effects in the dynamic version are temporally correlated, with information being transmitted between the time-steps. For instance, if a coefficient has a minimal shrinkage at time-step t , the prior for that coefficient in time-step $t + 1$ is likely to favour the minimal shrinkage as well.

Furthermore, the dynamic process specifies the majority of the hyperparameters used, which minimizes the hyperparameters selected by the user; see (2.25). Nevertheless, the user still needs to specify the value of α and β to select the respective dynamic process (see Table 2.1), which, in this case, must be $\alpha = \beta = \frac{1}{2}$.

Pólya-Gamma Process

When using these models, a key computational challenge is the heavy-tailed Z -distribution. Kowal, Matteson, and Ruppert [15] show that it is possible to use Gibbs sampling if the Z -distribution in (2.24) and in (2.25) is expressed by data augmentation using a Pólya-gamma variable.

A Pólya-gamma random variable ξ , with parameters $b > 0$ and $c \in \mathbb{R}$, denoted $\xi \sim PG(b, c)$, is equal in distribution to an infinite convolution of Gamma variables.

$$\xi \stackrel{d}{=} (2\pi^2)^{-1} \sum_{k=1}^{\infty} g_k \left(\left(k - \frac{1}{2} \right)^2 - \frac{c^2}{4\pi^2} \right)^{-1}, \quad (2.26)$$

$$g_k \stackrel{IID}{\sim} \text{Gamma}(b, 1).$$

Furthermore, the Pólya-gamma random variable allows to re-write the Z -distribution as a mean-variance scale mixture of Gaussian distributions

$$\begin{aligned} \eta_t | \xi &\sim N(\xi^{-1}(\alpha - \beta)/2, \xi^{-1}) \\ \xi &\sim PG(\alpha + \beta, 0) \end{aligned} \quad (2.27)$$

The data augmentation specified in (2.27) enables the construction of an efficient sampling algorithm ($O(T)$) for the log-volatilities h_t . This is possible due to the sampling algorithm for Pólya-gamma random variables,

proposed by Polson, Scott, and Windle [19], which is particularly efficient when $b = 1$, which is satisfied for the horseshoe prior where $\alpha = \beta = \frac{1}{2}$.

2.3 Markov chain Monte Carlo

Markov chain Monte Carlo (MCMC) methods are a general framework that allows one to approximate a target posterior distribution $P(\theta|y)$ by sampling; MCMC methods are typically used when direct sampling is challenging.

MCMC sampling samples from a Markov chain with a transition kernel designed to have a stationary distribution $P(\theta|y)$ ([20]). The Markov chain $(\theta^{(k)})$ is such that, as the number of iterations increases, the autocorrelated draws $\theta^{(k)}$ converge in distribution to $P(\theta|y)$; see Robert and Casella [20] for conditions. Thus, samples from the Markov chain can be used to approximate the target distribution.

There are several well-known algorithms that implement the general structure of MCMC; for this study, the Metropolis-Hastings (MH) algorithm and Hamiltonian Monte Carlo (HMC) were considered. The main difference between MCMC algorithms lies in the transition kernel used to construct the Markov chains.

2.3.1 Metropolis-Hastings

The general Metropolis-Hastings algorithm was already presented in Section 2.2.3; this section presents a specific proposal distribution, giving rise to the random-walk MH (RW-MH) algorithm. The next section presents a more sophisticated proposal based on ideas from physics.

The RW-MH algorithm uses a symmetric proposal distribution $Q(\theta^*|\theta^{(k-1)})$ centered around the current parameter $\theta^{(k-1)}$; the symmetry means that $Q(\theta^*|\theta^{(k-1)}) = Q(\theta^{(k-1)}|\theta^*)$. This approach induces a local exploration. In other words, the algorithm tends to propose new draws locally around the current parameter draw $\theta^{(k-1)}$. The most commonly used proposal distribution in RW-MH for parameters in $\mathbb{R}^p$ is $\theta^*|\theta^{(k-1)} \sim N(\theta^{(k-1)}, c\Sigma)$ for some covariance matrix Σ and scaling constant $c > 0$. The acceptance probability in the RW-MH becomes

$$\alpha = \min\left\{1, \frac{P(\theta^*|y)}{P(\theta^{(k-1)}|y)}\right\}, \quad (2.28)$$

which is simplified due to the symmetry of the proposal distribution.

The specific behaviour of this algorithm induces a major drawback: since RW-MH explores locally, it tends to take longer to explore the target posterior distribution, especially in high-dimensional settings where c needs to be set low to get a decent acceptance probability. Consequently, the RW-MH typically experiences slow mixing and produces highly correlated draws. These limitations motivate the introduction of more advanced algorithms, such as Hamiltonian Monte Carlo.

2.3.2 Hamiltonian Monte Carlo

The Hamiltonian Monte Carlo (HMC) algorithm uses concepts from physics (Hamiltonian dynamics) that help navigate the target posterior distribution faster than the MH algorithm, particularly in high-dimensional settings; see Gelman et al. [10] and Neal [17] for introductions to HMC. Traditional MCMC methods, such as MH and Gibbs sampling, operate on the parameter space of θ , the parameter vector of interest. The HMC algorithm instead operates on an expanded parameter space by introducing an auxiliary variable, *momentum* ϕ , and it updates both θ and ϕ jointly in an iterative fashion; θ and ϕ have the same number of elements. In this setting, θ is denoted as the position (the parameter of interest), whereas the auxiliary momentum ϕ allows the algorithm to explore the parameter space more efficiently.

The introduction of an extra variable transforms the parameter space from d to $2d$ dimensions; the log target distribution on this new parameter space can be defined by the Hamiltonian function - $H(\theta, \phi)$, as described in Neal [17]. The Hamiltonian function used in the HMC algorithm is typically specified as:

$$\begin{aligned} H(\theta, \phi) &= U(\theta) + K(\phi) \\ U(\theta) &= -\log(P(\theta|y)) \\ K(\phi) &= \frac{\phi^\top M^{-1} \phi}{2}, \end{aligned} \tag{2.29}$$

where H is the *total energy*, decomposed into the *potential energy* U , and the *kinetic energy* K , and M is a *mass matrix*. The partial derivatives of the Hamiltonian function are referred to as equations of motion; these equations control how θ and ϕ evolve over time:

$$\begin{aligned} \frac{d\theta_k}{dt} &= \frac{\partial H}{\partial \phi_k} \\ \frac{d\phi_k}{dt} &= -\frac{\partial H}{\partial \theta_k}, \end{aligned} \tag{2.30}$$

where time refers to the time from the initial push from the current parameter value $\theta^{(k-1)}$ to the terminal position at the proposed value θ^* . As suggested by Neal [17], the use of Hamiltonian function is justified by the properties of Hamiltonian dynamics, which are ideally suited to MCMC methods; The properties in question are reversibility, conservation of the Hamiltonian, and conservation of the volume in the parameter space.

Following the framework presented in Gelman et al. [10], the target distribution of HMC is the joint distribution of θ and ϕ

$$P(\theta, \phi|y) = P(\phi)P(\theta|y), \quad (2.31)$$

which clearly has $P(\theta|y)$ as the marginal distribution. The only relevant draws are the θ draws, and the ϕ -draws can therefore be discarded.

The momentum distribution $P(\phi)$ typically is a multivariate normal distribution with mean zero and covariance M , usually a diagonal mass matrix, and is assumed independent of θ . In addition, as suggested by Neal [17], the joint distribution can also be expressed as the canonical distribution, depending on the total energy:

$$P(\theta, \phi|y) \propto \exp(-H(\theta, \phi)). \quad (2.32)$$

From the Hamiltonian dynamics in (2.30) and the definition of the potential energy, we see that

$$-\nabla_{\theta}H(\theta) = \nabla_{\theta} \log P(\theta|y), \quad (2.33)$$

so that HMC relies on the gradient information of the log-posterior to propose trajectories of high probability density. The gradient information is used to update the momentum variable ϕ in the Hamiltonian dynamics, whereas the position variable θ is updated by the updated momentum variable ϕ . Consequently, the HMC algorithm can propose states that are not in proximity with the current state and still obtain a high acceptance probability, since the Hamiltonian dynamics conserves the total energy $H(\theta, \phi)$; this explains why HMC is so effective in high-dimensional parameter spaces.

In terms of computer implementation, it is important to note that Hamilton's equations usually do not admit a closed form solution, due to the complexity of the target distribution; generally, it is possible to compute an approximation using numerical methods. The method most frequently used to calculate the approximation is the leapfrog method, see for example Neal [17]. This method is based on discretizing time using a small step

size ϵ , and is defined by

$$\begin{aligned}
\tilde{\phi}^{(l)} &= \phi^{(l-1)} + \frac{\epsilon}{2} \frac{\partial \log P(y|\theta)P(\theta)}{\partial \theta} \Big|_{\theta^{(l-1)}} \\
\theta^{(l)} &= \theta^{(l-1)} + \epsilon M^{-1} \tilde{\phi}^{(l)} \\
\phi^{(l)} &= \tilde{\phi}^{(l)} + \frac{\epsilon}{2} \frac{\partial \log P(y|\theta)P(\theta)}{\partial \theta} \Big|_{\theta^{(l)}},
\end{aligned} \tag{2.34}$$

it is worth noting that the half-step for the momentum and the negation of the momentum at the final iteration (note that it was not presented in the equations for the sake of clarity) gives the reversibility property and also improves the numerical accuracy. The leapfrog steps in (2.34) are iterated L times for each proposal. A clear advantage of the leapfrog method is that it preserves the volume and is also reversible; as stated earlier, these two characteristics fit perfectly with MCMC methods. On the other hand, a disadvantage is that the method requires parameter (ϵ and L) tuning for good performance.

The HMC algorithm is based on two stages; in the first stage, new initial values of the momentum variables are randomly sampled (note that these samples are independent of the current values of the position variables). In the second stage, a proposed state (θ^*, ϕ^*) is generated using the Hamiltonian dynamics, discretized using the Leapfrog method. The proposed state is then accepted with probability:

$$\min[1, \exp(-H(\theta^*, \phi^*) + H(\theta, \phi))]. \tag{2.35}$$

If the proposed state is rejected, the current state remains unchanged. The option of accept/reject the proposed states was implemented to compensate for the numerical errors accumulated by the Leapfrog method.

The complete HMC algorithm is presented for the sake of completeness and clarity, following the framework presented in Neal [17]

HMC algorithm

0. Initialize $\theta^{(0)}$;
1. for t in $1, \dots, T$:
 - (a) Sample momentum: $\phi^{(t)} \sim N(0, M)$;
 - (b) Set $(\theta^{(t-1)}, \phi^{(t)})$ as current state;
 - (c) Leapfrog Method:
 - i. for l in $1, \dots, L$:

- A. $\phi^{(l)} = \phi^{(l-1)} + \frac{\epsilon}{2} \frac{\partial \log P(y|\theta)P(\theta)}{\partial \theta} \Big|_{\theta^{(l-1)}}$
- B. $\theta^{(l)} = \theta^{(l-1)} + \epsilon M^{-1} \phi^{(l)}$
- C. $\phi^{(l)} = \phi^{(l)} + \frac{\epsilon}{2} \frac{\partial \log P(y|\theta)P(\theta)}{\partial \theta} \Big|_{\theta^{(l)}}$
- (d) Set $(\theta^* = \theta^{(L)}, -\phi^* = \phi^{(L)})$;
- (e) Compute acceptance probability:
 - i. $\alpha = \min[1, \exp(-H(\theta^*, \phi^*) + H(\theta^{(t-1)}, \phi^{(t)}))]$
- (f) Draw $u \sim \text{Uniform}(0, 1)$:
 - i. if $u \leq \alpha$ set $\theta^{(t)} = \theta^*$
 - ii. if $u > \alpha$ set $\theta^{(t)} = \theta^{(t-1)}$

No-U-Turn Sampler

A drawback of the HMC algorithm is the necessity of pre-specify the algorithm's tuning parameters, such as L and ϵ . Since ϵ should be kept small to maintain a high acceptance probability, if L is chosen to be too small, there is a risk that the HMC will exhibit a random-walk behaviour. On the other hand, if L is too large, the trajectory may return to its starting point, which wastes computational resources. In the literature, a solution is suggested by Hoffman and Gelman [11], which presents the No-U-Turn Sampler (NUTS), an extension of the HMC algorithm that tunes the step size ϵ and the number of steps L automatically to obtain a pre-set average acceptance probability.

2.3.3 Divergences in Stan

Stan uses Hamiltonian Monte Carlo algorithms (HMC and NUTS) that simulate trajectories of the parameters in an expanded parameter space. The simulated trajectories are generated by a numerical approximation (Leapfrog algorithm) of the Hamiltonian dynamics and may occasionally deviate radically from the true Hamiltonian dynamics (Stan Development Team [23]); when this happens, it is referred to as a divergent transition; the Figure 2.2 illustrates an example of a divergent transition, following the framework presented in Neal [17].

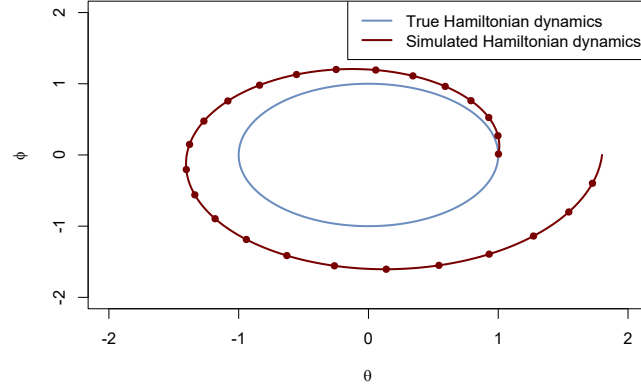


Figure 2.2: Example of a divergent transition

Divergent transitions tend to be more likely to occur when working with more complex models, such as hierarchical models, where the posterior distribution is typically highly curved and/or has a problematic geometry. When divergent transitions occur frequently, the inferred posterior distribution may not be reliable, since whenever a transition diverges, the positions present in that transition will not be selected as the next draw (Stan Development Team [23]). Consequently, the inferred posterior would be biased in favour of the regions with more favourable geometry, while under-representing the regions with less favourable geometry. Therefore, it is important to try to lower the divergent transitions as much as possible, ideally down to zero; the Figure 2.3 illustrates an example of joint posterior draws, while highlights the divergent transitions with red dots.

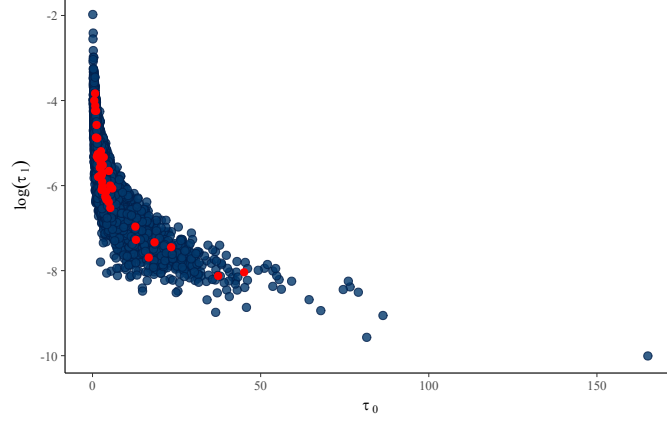


Figure 2.3: Joint posterior draws of $\log(\tau_1)$ and τ_0 (blue points) with starting points with divergent transitions as red points.

2.3.4 Effective Sample Size

In MCMC methods, the draws $\theta^{(k)}$ are typically autocorrelated; in general, draws that are closer in time tend to exhibit a stronger correlation. Therefore, the estimator for the Monte Carlo standard error of the inferred posterior should be modified. Since the samples are not independent, the error estimates may be underestimated or overestimated.

If the samples are independent and identically distributed (IID), the Monte Carlo standard error (MCSE) is proportional to $\frac{1}{\sqrt{N}}$; however, if the samples are dependent, the MCSE is proportional to $\frac{1}{\sqrt{N_{eff}}}$. Here N_{eff} is the *effective sample size*, which adjusts the nominal sample size N to take into account the autocorrelation [23]. The effective sample size can be defined as [23]

$$N_{eff} = \frac{N}{1 + 2 \sum_{k=1}^{\infty} \rho_k}, \quad (2.36)$$

where ρ represents the autocorrelation at lag k ,

$$\rho_k = \text{Corr}\left(\theta^{(n)}, \theta^{(n+k)}\right). \quad (2.37)$$

The ratio N/N_{eff} indicates how much information is lost as a result of the autocorrelation in the chain, and typically $N_{eff} < N$. This explains the previous comment that the MCMC error would be underestimated if the typically positive autocorrelation between the draws were not taken under consideration. Note that in the unlikely case where there is a negative correlation, $N_{eff} > N$, thus the error measurement will be overestimated.

3

Method

3.1 Research Strategy

This study compares the Gibbs sampling for the dynamic shrinkage process developed in Kowal, Matteson, and Ruppert [15] implemented in the R package `dsp` with a Stan implementation of the same model with posterior samples obtained by the HMC algorithm in R. Comparisons are made on both synthetic and real data. The sampling methods are compared along several dimensions: accuracy of the resulting posterior with respect to the data generating process, sampling efficiency, and computing time. The indicator used to assess the sampling efficiency was the effective sample size N_{eff} per minute; the coda package (version 0.19.4.1) was used to compute the N_{eff} ; more details on this package are available in the official documentation.

A Stan implementation of the model can be done in many different parametrizations. The Stan model was therefore carefully chosen from extensive testing in a preliminary phase, before comparing it to the Gibbs sampling version. Based on this assessment, a parametrization of the model was selected using the number of divergences and computation time as the primary criteria.

The parametrization of the model used for Gibbs sampling follows the one implemented in the `dsp` package. However, the package was cloned and modified to allow for a modified offset in the parameter innovations, since Fagerberg, Villani, and Kohn [6] show that this offset can have a large

effect on the posterior results. Details about the `dsp` package are available in the official documentation and in Section 3.2.4.

After the selection of parametrizations of the models, the Stan and DSP implementations were compared on the same synthetic data to ensure a controlled environment.

Real-world data experiments were conducted to assess the practical applicability of both implementations and to contrast the posterior distributions for interpretable parameters in a real-world setting.

By following this approach, a proper evaluation of the models is ensured, as they are tested and compared in both a controlled and a realistic environment.

3.2 Research Method

This section presents the details about the setup of the simulation experiments on synthetic data and how the data in the real-world application were collected.

The details of the implementations of the Stan and Gibbs sampling models are explained, along with the parametrization chosen for both. In addition, the techniques used to reduce the divergent transitions in the Stan implementation are also presented.

All code and datasets used for data generation, model implementation, and analysis are publicly available in a GitHub repository.

3.2.1 Simulation experiments

The synthetic dataset was generated using $T = 500$ data points and three regressors, one of which was used as an intercept. The regressors were generated from a standard normal distribution. Whereas, the regression coefficients were generated based on three specific functions: sinusoidal, step, and quadratic functions; these functions can be formulated as:

$$\begin{aligned}\beta_{1,t} &= \sin\left(\frac{10t}{T}\right) \\ \beta_{2,t} &= \begin{cases} -0.5, & \text{if } \frac{t}{T} < 0.5, \\ 0.5, & \text{otherwise} \end{cases} \\ \beta_{3,t} &= \left(\frac{t}{T}\right)^2,\end{aligned}\tag{3.1}$$

where t denotes the time-step and $T = 500$. The evolution of the coefficients is represented in Figure 3.1. These different patterns were chosen to assess the flexibility of the Stan and Gibbs sampling implementations. More specifically, to determine whether the implementations were capable of adapting to such diverse patterns.

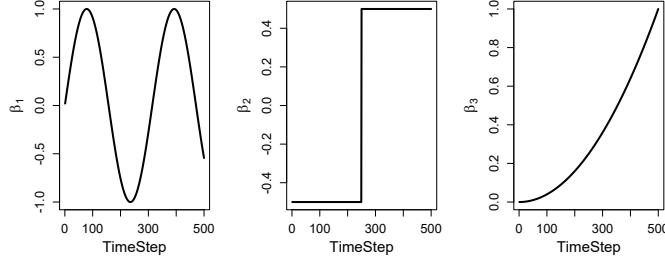


Figure 3.1: Evolution of the regression coefficients

In addition, an additive noise term ϵ was also included in the process; it was generated from a normal distribution, with zero mean and standard deviation of 0.45.

3.2.2 Real-world data experiments

The Stan and Gibbs sampling implementations were tested using the Capital Asset Pricing Model (CAPM), initially developed by, among others, Sharpe [22] and Lintner [16]. The model will be briefly introduced for clarity reasons. CAPM is a well-known model in the field of financial analysis that assess the performance of stocks; more specifically, it explains the relationship between systematic risk and expected return on an asset. The classical CAPM follows the model:

$$y_t = \alpha + \beta z_t + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2), \quad (3.2)$$

where $y_t = r_t - r_t^f$, with r_t as the return of the asset of interest at time t and r_t^f is the return of a risk-free asset; the covariate is defined as $z_t = r_t^M - r_t^f$, where r_t^M is the return of the market portfolio at time t . Therefore, y_t can be seen as the excess return of the stock at time t , while z_t can be seen as the excess return of the market at time t . Furthermore, the α indicates the excess returns not explained by the market, while β quantifies the sensitivity of the asset of interest to changes in the market return. The β parameter is important since:

- $\beta > 1 \rightarrow$ intensifies changes in market return and is regarded as an aggressive investment;
- $\beta < 1 \rightarrow$ diminishes changes in market return and is viewed as a conservative investment;

The classical CAPM model is a static regression with constant α and β , but it seems reasonable to allow the risk characteristics of a firm to change over time, as shown in Ang and Kristensen [1]. This suggests that the original CAPM model can be naturally extended to a TVP regression framework, thereby fitting the general framework required to use the samplers under study.

The data used are from Berndt [2, chapter 2] and consist of 17 stock returns, one market return, and one risk-free asset return. The data are monthly returns from January 1978 to December 1987, totalling 120 data points ($T = 120$). The response variables and the regressors were computed as suggested above. The selected assets of interest in this study are the IBM stock (technology), DELTA stock (aviation), and MOBIL stock (oil). This selection allows the exploration of different market sectors, consequently exposing the samplers under study to different economic behaviours.

It is worth highlighting that the dataset was already processed, therefore no data cleaning techniques were necessary, and the original data is available in a public GitHub repository.

A natural extension of the CAPM was used to evaluate the samplers in a more complex setting. The model in question is the Fama-French three factor model, presented in Fama and French [7]; note that for clarity reasons, the Fama-French three factor model will be denoted as Fama-French model. This model extends the CAPM by introducing two extra explanatory variables (known as factors in financial literature):

- $SMB \rightarrow$ Size factor - difference between the average returns of small stock portfolios and big stock portfolios;
- $HML \rightarrow$ Value factor - difference between the average returns of value portfolios and growth portfolios;

Moreover, the Fama-French model can be formulated as follows:

$$y_t = \alpha + \beta_1 z_t + \beta_2 SMB + \beta_3 HML + \epsilon_t, \quad \epsilon_t \sim N(0, \sigma_\epsilon^2), \quad (3.3)$$

where z_t is the already known excess market return variable from the CAPM. Similar to the CAPM, the Fama-French model was originally developed with

constant parameters. However, Ang and Kristensen [1] also presented findings that suggested that the parameters of the Fama-French model are time-varying. Therefore, as in CAPM, the Fama-French model can be naturally extended to a TVP regression framework.

The data used on the Fama-French experiment is similar to the data used on the CAPM experiment; the only difference being that the data for the two new factors (SMB and HML) were collected from Kenneth R. French's website. It is worth highlighting that the original data was already processed, therefore the only data cleaning technique used was identifying missing values. Since in Kenneth R. French website is mentioned the possibility of existing missing values, although they have a value associated to them (-99.99 and -999). Nevertheless, in the selected data for this study, no missing values were detected.

3.2.3 Stan Implementation

The implementation in Stan follows a general configuration very similar to the one presented in 2.25. The only differences are the distribution chosen for the priors of ξ and σ_ϵ^2 . In the original model 2.25, an implementation of a Pólya-Gamma distribution for ξ is suggested; Stan does not support that distribution, so an approximation using a Log-Normal distribution was developed:

$$\xi_{j,0} \sim \text{Log} - \text{Normal}(\mu = -1.683977, \sigma = 0.7891323) \quad (3.4)$$

The parameters of the Log-Normal distribution were computed from the empirical mean and standard deviation of a simulation involving 10000 draws of $\xi \sim PG(0, 1)$. This approximation can be visualized in Figure 3.2.

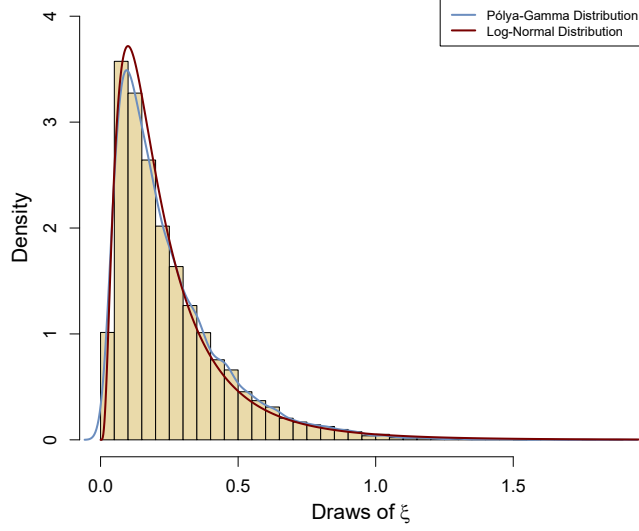


Figure 3.2: Distribution of ξ draws

Furthermore, the original model also suggested the implementation of a Jeffreys prior on the observation error variance σ_ϵ^2 . This prior can also be expressed in standard deviation form, as demonstrated in Villani [24]:

$$\sigma_\epsilon \propto \frac{1}{\sigma_\epsilon}. \quad (3.5)$$

This last formulation is more useful since Stan works directly with the standard deviation. Nevertheless, the selected prior for the observation error standard deviation σ_ϵ was a Scaled Inverse Chi-Square distribution, defined as:

$$\sigma_\epsilon \sim \text{ScaledInvChiSquare}(\nu, \sigma). \quad (3.6)$$

Although the Jeffreys prior is a non-informative prior, it still exhibits some undesirable characteristics, such as the possibility of generating improper posterior distributions. Consequently, the Scaled Inverse Chi-Square distribution was selected, a weakly informative prior that ensures a proper posterior distribution and numerical stability. This new prior has a similar behaviour as the Jeffrey prior, as illustrated by Figure 3.3. This similarity holds only when both hyper-parameters take positive values near zero; in this study, $\nu = \sigma = 0.00001$ was adopted.

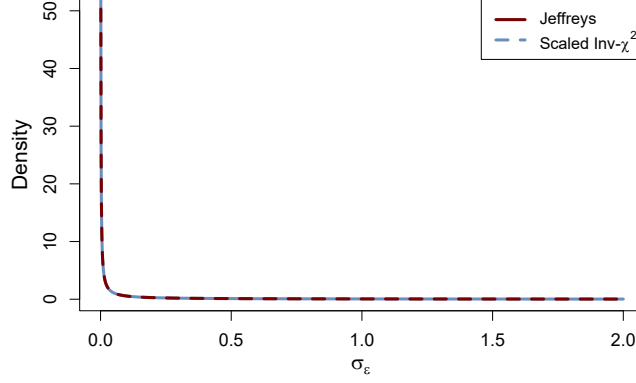


Figure 3.3: Comparison between Jeffreys prior and Scaled $\text{Inv-}\chi^2$ prior

Note that both densities were normalized over a bounded interval (0.0001, 2), which allows for a direct comparison of their shapes.

Stan's implementations on synthetic data follow exactly the model described above. Although in the experiment with real-world data, two changes were made, following the framework presented in Kowal, Matteson, and Ruppert [15]. The observation error variance followed a stochastic volatility (SV) model, as specified by Kastner [13]:

$$\begin{aligned}
 \sigma_{\epsilon,t}^2 &= \exp(g_t) \\
 g_t | g_{t-1}, \mu, \phi, \sigma_\eta &\sim \text{N}(\mu + \phi(g_{t-1} - \mu), \sigma_\eta^2) \\
 g_0 | \mu, \phi, \sigma_\eta &\sim \text{N}\left(\mu, \frac{\sigma_\eta^2}{1 - \phi^2}\right) \\
 \mu &\sim \text{N}(0, 100) \\
 \frac{\phi + 1}{2} &\sim \text{Beta}(5, 1.5) \\
 \sigma_\eta^2 &\sim \text{Gamma}\left(\frac{1}{2}, \frac{1}{2}\right).
 \end{aligned} \tag{3.7}$$

Furthermore, the shrinkage parameters were also affected; instead of defining the shrinkage with three parameters $(\tau_0, \tau_j, \lambda_{j,t})$, both τ_0 and τ_j were merged, becoming a single global parameter again. Thus, the priors for the model also change; the single change is defined below:

$$\tau \sim C^+\left(0, \frac{1}{\sqrt{T}}\right). \tag{3.8}$$

In addition, all details of the initialization of the models will be specified, to ensure maximum reproducibility, as specified in Table 3.1.

Data type	Chains	Warmup Iteration	Sampling Iterations	Seed	adapt_delta	max_treedepth
Synthetic	1	1000	12000	123	0.995	15
Real-world data	1	1000	12000	123	0.995	15

Table 3.1: Sampling settings (Stan)

Note that every Stan implementation used R software (version 4.51), the `cmdstanr` package (version 0.9.0), and was executed in the Tetralith cluster.

Reducing divergences in Stan

In this study, it was used some known techniques to lower the divergent transitions, since after building the model proposed by Kowal, Matteson, and Ruppert [15], several divergent transitions occurred. The two techniques that produce better results were non-centered distributions and inverse transform sampling. These techniques allow to transform the geometry of the posterior for a more favourable one, while maintaining mathematical equivalence. Now, it will be presented both:

Non-centered Distribution

Assume $X \sim N(\mu, \sigma^2)$.

X may also be expressed as: $X = \mu + \sigma * Z$, where $Z \sim N(0, 1)$.

Inverse transform sampling

Assume X as a random variable and F_X as the cumulative distribution function (CDF).

1. Generate a random number u , where $u \sim \text{Uniform}(0, 1)$;
2. Find $F_X^{-1}(u)$ (generalized inverse);
3. Compute $X' = F_X^{-1}(u)$;
4. X' will follow the same distribution as X ;

In addition, the step size in the HMC sampler was set to a value lower than the default, achieved by increasing the `adapt_delta` - target average proposal acceptance probability in the NUTS algorithm. This does not influence the geometry of the posterior distribution, but helps the sampler to navigate the posterior more robustly by using a more careful discretization of the Hamiltonian dynamics. Note that, as the value of `adapt_delta` increases, the computational time will typically increase as well, as represented by Figure 3.4.

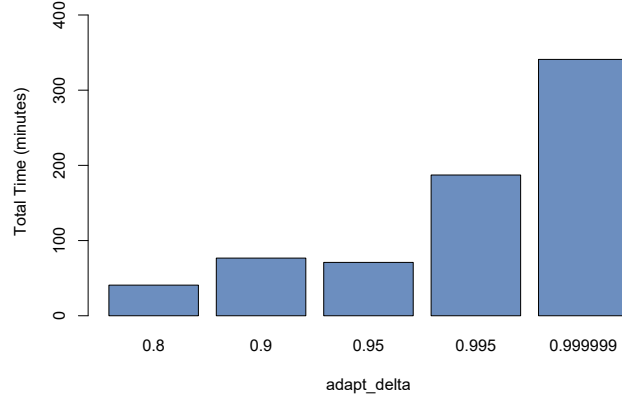


Figure 3.4: Trade-off between runtime and adapt_delta

3.2.4 Gibbs sampling implementation

The Gibbs sampling implementation in this study is almost identical to the original dsp package. There was a minor change on the sampler of the log-volatilities; originally, the package adds an offset to the calculations to ensure numerical stability, as presented in equation 3.9.

$$\log((\omega_t + \text{offset})^2) = h_t + \zeta_t. \quad (3.9)$$

This offset typically adapts its value depending on the scale of the data. In this study, this offset was defined as a fixed value ($2.220446\text{e-}12$), typically much lower than the default in the dsp package. The main objective of this change is for the model to produce a more stable and smoother posterior distribution since a too large offset will counteract the dsp’s ability to produce constant parameter evolutions, as highlighted in Fagerberg, Villani, and Kohn [6].

Moreover, the implementation on real-world data also applied a stochastic volatility model to the observation error variance, as suggested by Kowal, Matteson, and Ruppert [15]; this model followed the framework described in equation 3.7.

All details regarding the models used in the thesis are specified in Table 3.2.

Data type	Familiy	Model	D	evol_error	obsSV	nsave	nburn
Synthetic	gaussian	regression	1	DHS	const	12000	1000
Real-world data	gaussian	regression	1	DHS	SV	12000	1000

Table 3.2: Sampling settings (Gibbs)

All implementations used the R software (version 4.5.1) and `dsp` package (version 1.2.0 or 1.2.0.2); note that the version 1.2.0 represents the original `dsp` package, while the version 1.2.0.2 represents the modified package with the fixed offset. This package is publicly available in a GitHub repository.

4

Results

This section presents the results obtained in this study. Starting with an individual analysis of the Stan and Gibbs sampling implementations and concluding with a comparison of the final parametrisations of both.

For the sake of clarity, not all parameters were presented in this section, since not all of them are relevant to this discussion. Nevertheless, the remaining plots can be consulted in a public GitHub repository.

4.1 Selecting the Stan model and HMC tuning

As discussed in Section 3, a number of experiments were conducted to select the parametrisation of the Stan model for this study; the main results will be presented below.

One of the noteworthy experiments involved varying the `adapt_delta` parameter, the target average proposal acceptance probability; Figure 4.1 illustrates the trade-off between the total runtime of the Stan model and the number of divergences. The effects of `adapt_delta` are as expected: when `adapt_delta` is increased, the compute time increases and the number of divergences decreases. It is worth noting that the decreases in the number of divergences beyond the 0.995 threshold are marginal; there is only a decrease of two divergences when comparing parametrisations with an `adapt_delta` of 0.995 and 0.999999. Nevertheless, the difference in computational time is not marginal; between the two parametrisations, there is a difference of 272.58 minutes.

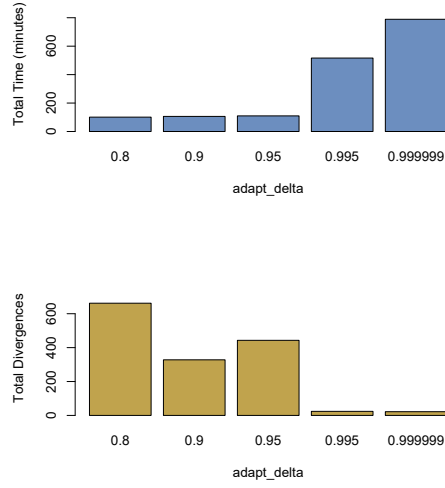


Figure 4.1: Trade-off between runtime (top) and HMC divergences (bottom)

The impact of the divergences on the inferred posterior distribution of the parameters was also examined; the conclusion of that study was that, although there is a clear difference on the number of divergences for each parametrisation, the effect of this on the posterior inference is not so evident, as illustrated by Figure 4.2; the posterior medians and the 95% point-wise highest posterior density (HPD) credible intervals are very similar, regardless the `adapt_delta` selected. Note that the thicker lines represent the posterior medians; the colour scheme is represented below:

Colour	<code>adapt_delta</code>
Blue	0.8
Yellow	0.9
Red	0.95
Green	0.995
Purple	0.999999

Table 4.1: Colour scheme

Furthermore, the **black line** represents the true parameter path. The HPD credible intervals are represented by bands (only for the parametrization with 0.8 of `adapt_delta`) and by thinner lines; the colour scheme is specified in Table 4.1.

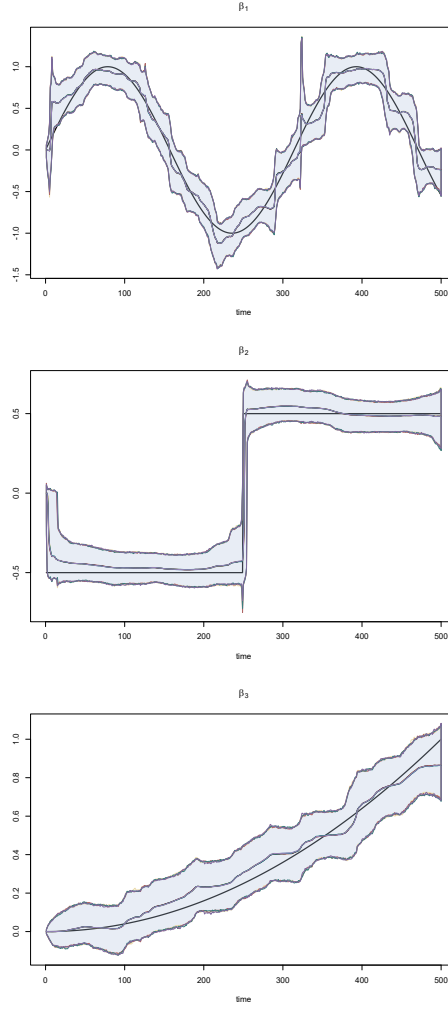


Figure 4.2: Inferred posterior medians for the three β (thick line) and 95% HPD credible intervals (shaded and thin lines) using the Stan implementation with different `adapt_delta` for the NUTS tuning of the HMC sampler. The results for each `adapt_delta` are represented by a colour, as specified in Table 4.1.

The hyperparameter posteriors shown in Figures 4.3 - 4.5 also imply that the divergences do not highly impact the inferences for these static parameters, at least for this dataset. The colour scheme follows Table 4.1.

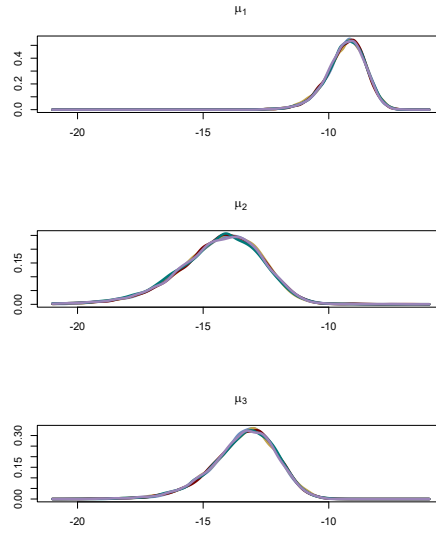


Figure 4.3: Inferred posterior densities for μ (Stan), where each μ_k is specific to β_k . Each parametrization of the Stan model is represented by a colour, as specified in Table 4.1.

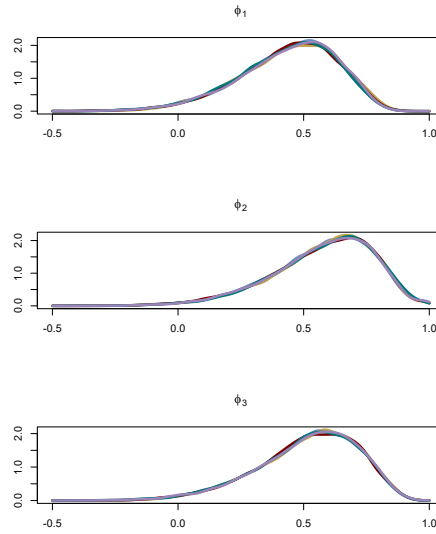


Figure 4.4: Inferred posterior densities for ϕ (Stan), where each ϕ_k is specific to β_k . Each parametrization of the Stan model is represented by a colour, as specified in Table 4.1.

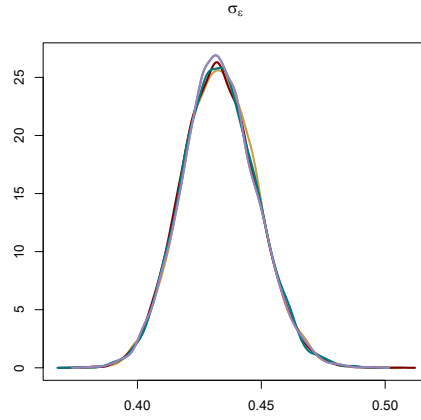


Figure 4.5: Inferred posterior density for σ_ϵ (Stan). Each parametrization of the Stan model is represented by a colour, as specified in Table 4.1.

In terms of prediction, the divergences do not appear to have an extensive impact as well, since all parametrisations inferred a similar distribution for the response variable (y) in Figure 4.6: The posterior predictive distributions closely match the empirical distribution of the data. Note that the colour scheme in Figure 4.6 follows the scheme presented in Table 4.1 and that the purple line overlaps the others' posterior predictive distributions, indicating that they are really similar.

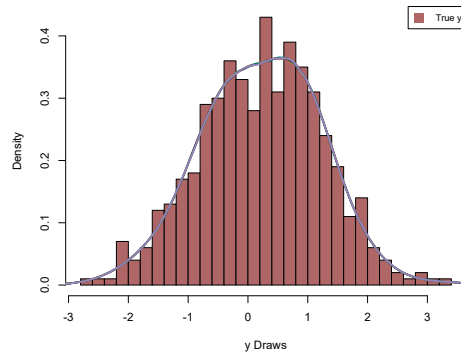


Figure 4.6: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (solid lines). The line colours indicate the `adapt_delta` tuning parameter of the NUTS algorithm, as specified in Table 4.1.

The NUTS tuning with `adapt_delta` set to 0.995 was selected, since it was the most balanced in terms of number of divergences (24 divergences) and computation time (516.47 minutes). Note that, in this case, the divergences do not affect dramatically the inference of the model, nevertheless, in general, the aim is to lower the divergences as much as possible, as discussed in Section 2.3.3. Thus, the parametrization with an `adapt_delta` set to 0.995 was chosen over the faster parametrizations, such as the one with an `adapt_delta` set to 0.8 (run time: 101.03 minutes and divergences: 662).

4.2 Selecting offset in the Gibbs sampling algorithm

As outlined in Section 3, the offset used in the original `dsp` package has a significant impact on the posterior results. Therefore, understanding the extent of this offset is crucial. To this end, the posterior results from two models will be investigated: one using the original `dsp` package and the offset used there, and the other using the modified `dsp` package where the offset can be chosen freely.; for this study, the offset was fixed at $2.220446e-12$. The modified `dsp` package is referred as DSP*.

As illustrated by Figure 4.7, both posteriors from both models follow reasonably well the true evolution, suggesting that the offset may not have a significant impact. The posteriors for β_1 are also very similar. However, the DSP* model presents a posterior with less variability and greater smoothness, for the parameters β_2 and β_3 ; the `dsp` model produces wider HPD credible intervals (see Table 4.2), and the posterior median tends to fluctuate more around the true parameter path. This indicates that the offset has a significant impact on the variability and smoothness of the posterior distribution.

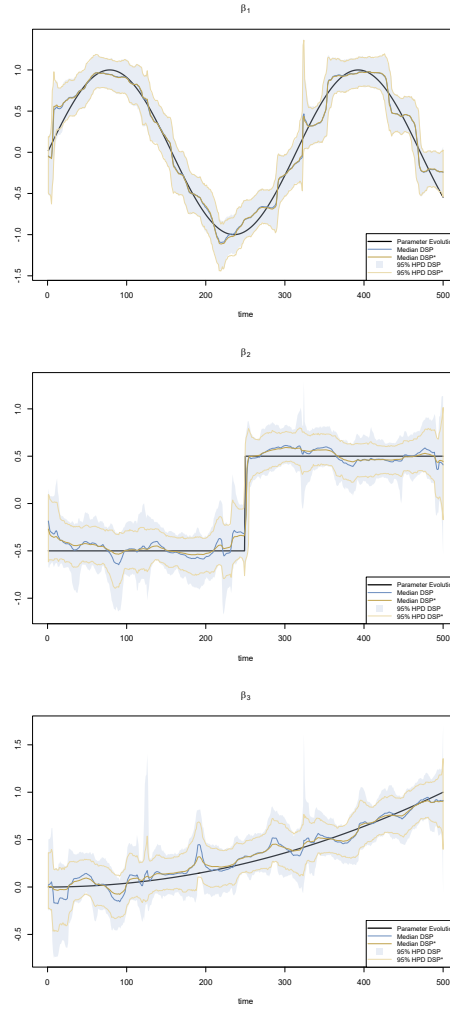


Figure 4.7: Inferred posterior medians for the three β (thick line) and 95% HPD credible intervals (shaded and thin lines) using the Gibbs implementation with different offsets. Each parametrisation is represented by a colour, as shown in the legend.

Parameter	Model	
	DSP	DSP*
β_1	0.52	0.51
β_2	0.57	0.41
β_3	0.56	0.42

Table 4.2: Mean of the width of the HPD credible intervals (Gibbs)

On the other hand, for predictive inference for the response variable (y), the type of the offset seems to be irrelevant, since both models produce a very similar posterior predictive distribution, as shown by Figure 4.8. In addition, both distributions closely approximate the empirical distribution of the response variable.

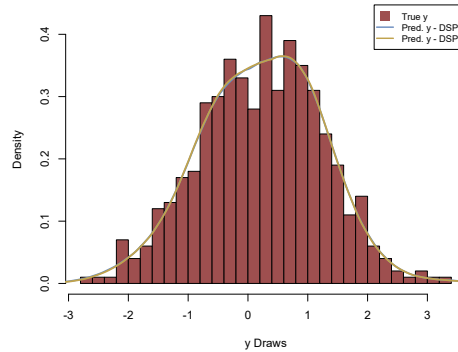


Figure 4.8: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from both Gibbs sampling parametrisations. The line colours indicate the parametrisation used, as shown in the legend.

In summary, the chosen offset is the one in the modified DSP*, as the posterior is smoother and less variable, particularly for β_2 and β_3 . Note that from this section onwards, the DSP* model will be referred to simply as DSP model.

4.3 Comparing the HMC and Gibbs sampling results

This section compares the results from the selected Stan implementation of the model sampled with HMC with the results from the Gibbs sampling in the dsp package with the modified offset. The Stan implementation will for simplicity, be referred to as the *Stan model* and the Gibbs sampling implementation as the *DSP model*, to reflect the slightly different parametrizations and offsets. This section will be divided in two sections: firstly, the results from the study on synthetic data will be addressed; and secondly, the results of the study on the real-world dataset will be presented.

4.3.1 Synthetic data

As shown in Figure 4.9, both posteriors track relatively well the true values of the parameters. However, the Stan posterior tends to be smoother and more stable than the DSP posterior; the DSP posterior presents, on average, a wider HPD credible interval (see Table 4.3) and tends to fluctuate more around the true parameter evolution. Note that the Stan model does not use an offset, which may explain the greater smoothness and lower variability of the posterior distribution. As in the last section, the posteriors of the parameter β_1 are very similar.

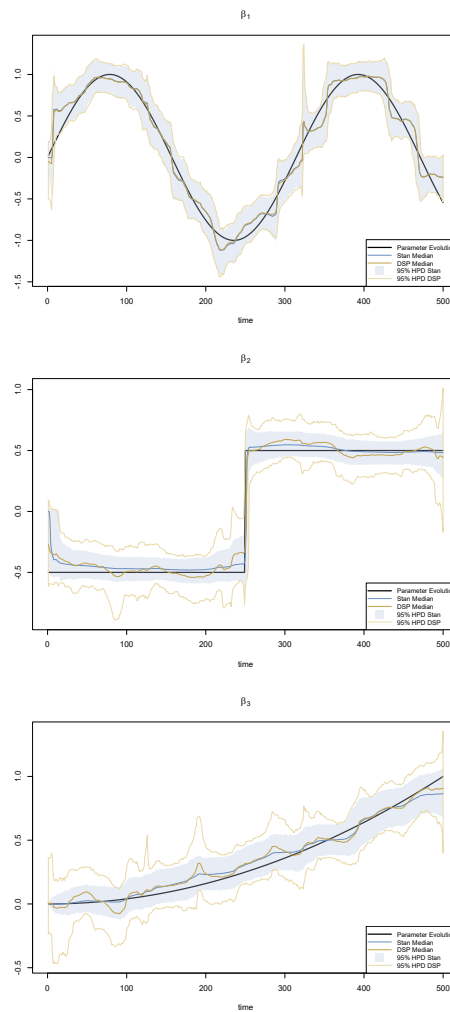


Figure 4.9: Inferred posterior medians for the three β (thick line) and 95% HPD credible intervals (shaded and thin lines) using the Stan and the DSP model. Each model is represented by a colour, as shown in the legend.

Parameter	Model	
	Stan	DSP
β_1	0.50	0.51
β_2	0.24	0.41
β_3	0.27	0.42

Table 4.3: Mean of the width of the HPD credible intervals (Stan vs DSP model)

In terms of computational time, the DSP model clearly has the upper-hand, as suggested in Table 4.4. Nevertheless, this indicator alone is not strong enough; a better approach is to analyse the N_{eff} per minute, since this new indicator allows one to understand the exact number of effective independent samples per minute, i.e., the sampling efficiency that takes into account both the MCMC efficiency and the compute time; see the information on N_{eff} in Section 2.3.4. As illustrated by Figure 4.10, the DSP model also has a clear advantage in terms of sample efficiency; while the Stan model has an N_{eff} per minute between approximately 3 and 25, the DSP model has an N_{eff} per minute between approximately 100 and 3000. Even if the DSP model is compared with the fastest parametrization of the Stan model, with N_{eff} per minute between approximately 10 and 120, the DSP is clearly more efficient in sampling. Therefore, the DSP approach is faster and samples more efficiently than the Stan approach. It is important to note that Stan is written in compiled C++ code, while the dsp package is written using a mix of R and C++. A full C++ implementation of the dsp package would be much faster.

	Stan 0.995	Stan 0.8	DSP
Time	516.47	101.03	3.92

Table 4.4: Total runtime (in minutes) (Stan vs DSP model)

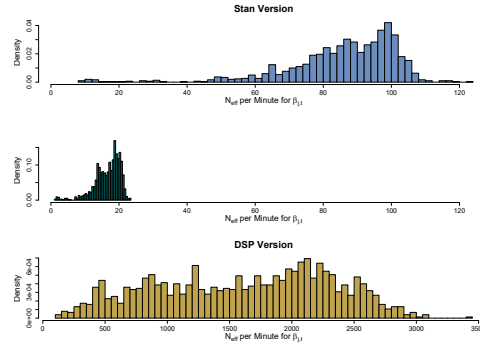


Figure 4.10: Distribution of N_{eff} per minute. The histograms for the Stan model represent the fastest tuning (`adapt_delta` - 0.8) and the selected tuning (`adapt_delta` - 0.995), and follow the colour scheme presented in Table 4.1. Note the widely different scales on the horizontal axes in the graphs.

Furthermore, in terms of prediction, both models produce a similar distribution for the response variable (y), despite the differences in computational efficiency; moreover, both distributions are similar to the true distribution, as shown in Figure 4.11. Therefore, despite their differences, both models demonstrate good predictive performance.

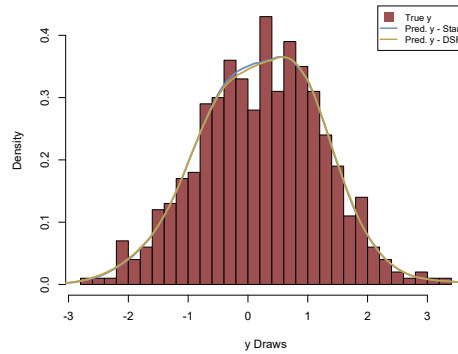


Figure 4.11: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (blue line) and Gibbs (yellow line).

4.3.2 Real-world data

In this section, the results from the real-world data experiments will be presented. As previously mentioned, this experiment was initially built around the CAPM model, since its framework allows for the implementation of the Stan and DSP models. Subsequently, the experiment was extended to a

more complex model, the Fama-French model, in order to understand the impact of increased complexity on the Stan and DSP models' performance.

The assets that were selected for this study are stocks of different economic sectors: IBM (technology), DELTA (aviation), and MOBIL (oil). This broad selection exposes the Stan and DSP models to different dynamics and economic behaviours.

This section will be structured into two main sections, each one focusing on the results of a specific experiment (CAPM and Fama-French). Furthermore, each main section will be divided into three subsections, corresponding to the three response variables (IBM, DELTA, and MOBIL).

CAPM results

IBM stock As shown in Figure 4.12, the Stan and DSP models produced a similar posterior distribution; for both parameters, the posterior medians follow the same pattern: the α medians fluctuate around the value zero, while the β medians show an increasing trend. Nevertheless, on average, the posterior for the β parameter presents a wider HPD interval for the DSP model, as reported in Table 4.5.

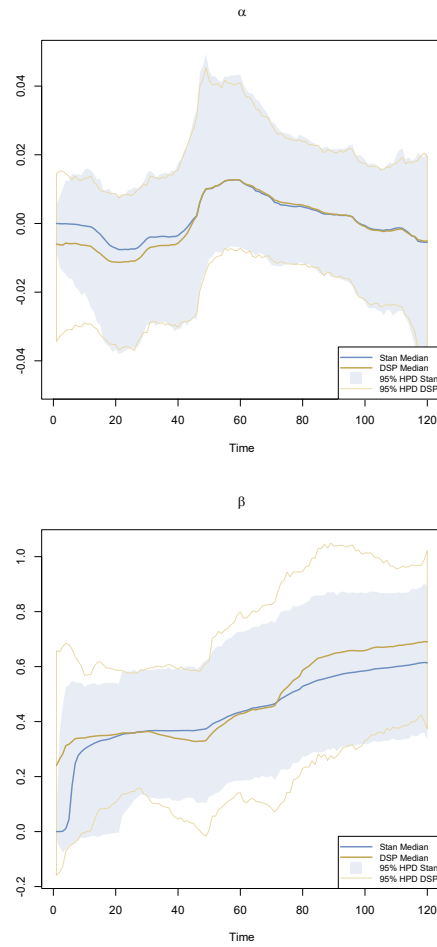


Figure 4.12: Inferred posterior over time for α_t (top) and β_t (bottom) in the CAPM model for the IBM stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

Parameter	Model	
	Stan	DSP
α	0.04	0.04
β	0.53	0.63

Table 4.5: Mean of the width of the HPD credible intervals (IBM stock - CAPM)

Moreover, for the observation error standard deviation σ_ϵ , the Stan and DSP models strongly agree on the posterior distribution. As illustrated by Figure 4.13, both posterior medians fluctuate around 0.05, and HPD credible intervals have similar widths.

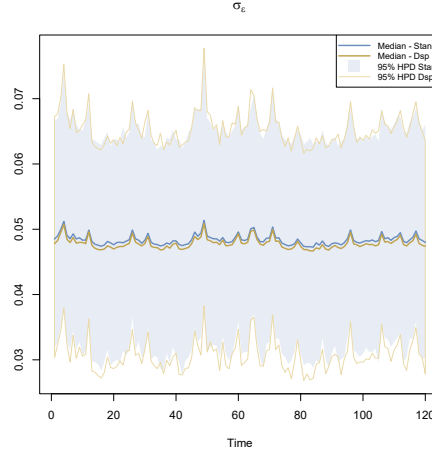


Figure 4.13: Inferred posterior over time for σ_ϵ (SV process) and 95% HPD credible intervals. The thicker lines represent the posterior medians, while the shaded area and thinner outer lines represent the credible intervals. Each model is represented by a colour, as shown in the legend.

Considering the total run time, the DSP model presents a much faster sampling algorithm; the run time is 4.78 minutes - less than half of Stan's computing time, as presented in Table 4.6. In addition, the DSP model shows a more variable N_{eff} per minute, ranging from approximately 80 to 2700. Whereas the Stan model concentrates the values more closely, ranging from approximately 204 to 850, as specified in Figure 4.14. Therefore, in contrast with the synthetic data experiment, it is not possible to claim that the DSP sampling is more efficient, since in some instances the DSP model demonstrates a lower N_{eff} per minute than Stan. The reason for the differences in N_{eff} per minute compared to the simulated data most likely comes from the much smaller sample size with only $T = 120$ time periods in the data.

Model	Stan	DSP
Time	13.40	4.78

Table 4.6: Total runtime (in minutes) (IBM stock - CAPM)

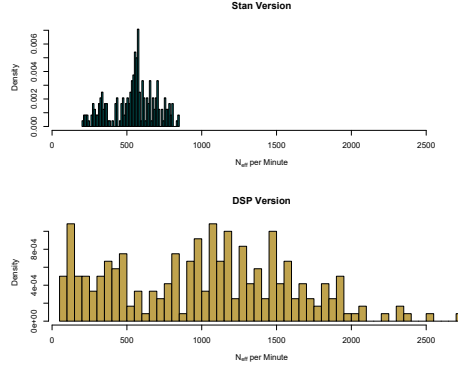


Figure 4.14: Distribution of N_{eff} per minute. The histogram for the Stan model represent the selected tuning (adapt_delta - 0.995), and follow the colour scheme presented in Table 4.1. Whereas the histogram for the DSP represents the implementation with the modified offset.

Despite the differences in the computation time and sampling efficiency, the Stan and DSP models present similar predictive power, as shown in Figure 4.15. Nevertheless, they fail to capture the central peaks of the distribution of the response variable (y) and part of the right tail.

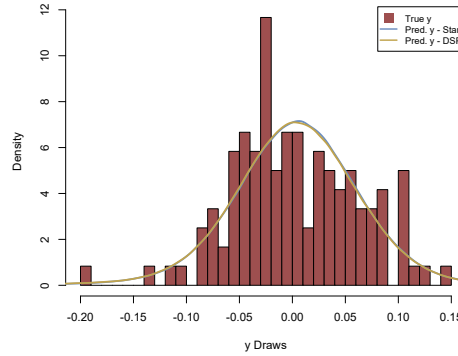


Figure 4.15: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (blue line) and Gibbs (yellow line).

DELTA stock As in the IBM stock, the Stan and DSP models inferred similar posterior distributions. The parameters medians also behave in the same way: the median of α tends to fluctuate around zero, while the median of β shows an upward trend; note that for the Stan model, the median of β is smoother, as displayed in Figure 4.16. Nevertheless, on average, the

posterior of β generated by the DSP model demonstrates a wider HPD credible interval, as presented in Table 4.7.

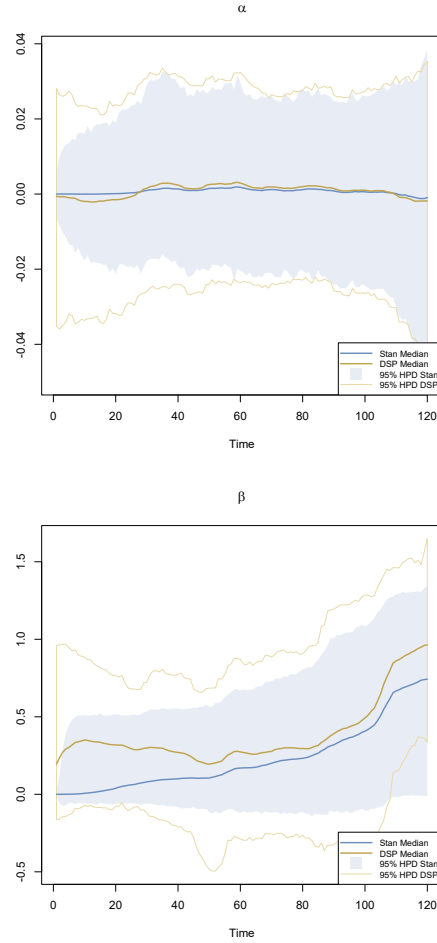


Figure 4.16: Inferred posterior over time for α_t (top) and β_t (bottom) in the CAPM model for the DELTA stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

Parameter	Model	
	Stan	DSP
α	0.05	0.06
β	0.84	1.16

Table 4.7: Mean of the width of the HPD credible intervals (DELTA Stock - CAPM)

The observation error standard deviation σ_ϵ clearly exhibits very similar posteriors distributions, as shown in Figure 4.17; the posterior medians are centered around 0.09, and even the spikes of volatility are matched. In addition, in terms of uncertainty, they are very similar as well.

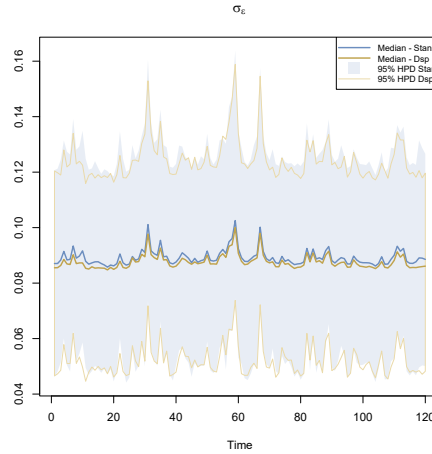


Figure 4.17: Inferred posterior over time for σ_ϵ (SV process) and 95% HPD credible intervals. The thicker lines represent the posterior medians, while the shaded area and thinner outer lines represent the credible intervals. Each model is represented by a colour, as shown in the legend.

The computation time is much closer than in previous experiments, with a difference of 3.51 minutes; however, the DSP model is still faster, as displayed in Table 4.8. The distribution of N_{eff} per minute shows the usual spread out histogram for the DSP model, with values approximately ranging from 462 to 3292. However, the Stan model presents a histogram more spread than usual, with almost two separate blocks, indicating that, in some instances, the sampling was more efficient. Perhaps this suggests that some areas of the posterior distribution were more problematic to sample. The N_{eff} per minute for the Stan model ranges approximately

from 365 and 1725. In this case, the DSP model appears to have an advantage in sampling efficiency, as its minimum value of N_{eff} per minute is higher than Stan's and even reaches higher values consistently, as illustrated by Figure 4.18.

Model	Stan	DSP
Time	6.96	3.45

Table 4.8: Total runtime (in minutes) (DELTA stock - CAPM)

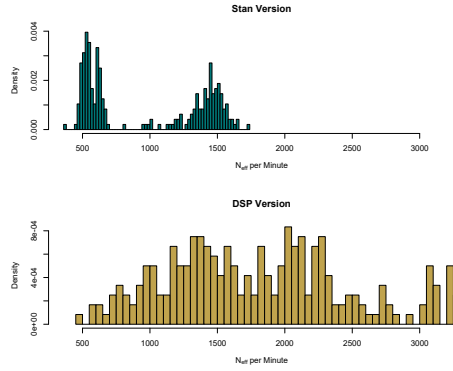


Figure 4.18: Distribution of N_{eff} per minute. The histogram for the Stan model represent the selected tuning (`adapt_delta` - 0.995), and follow the colour scheme presented in Table 4.1. Whereas the histogram for the DSP represents the implementation with the modified offset.

For a better understanding of the bimodal distribution for Stan in Figure 4.18, a plot showing the trajectories of the N_{eff} over time will be presented (Figure 4.19). This plot may explain the behaviour of the Stan model's histogram, since it shows that the α parameter has an average N_{eff} of 10000 over time, while the β parameter has an average N_{eff} of 4000 over time. Given that the average N_{eff} of both parameters is very distant, it is likely that each block represents the sampling efficiency of each parameter. In addition, it could also be inferred that the posterior distribution of the β parameter may have a more challenging geometry compared to the posterior of the α parameter.

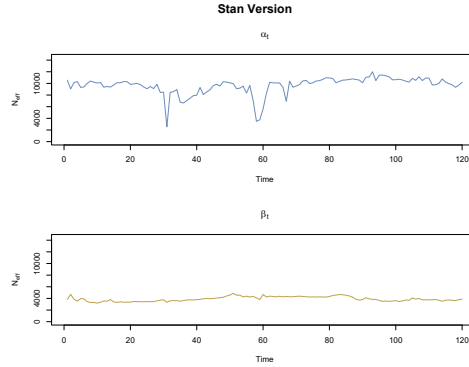


Figure 4.19: Trajectory of N_{eff} over time from the Stan model. Each parameter is represented by a colour: the blue colour represents the α_t and the yellow colour represents the β_t .

Once again, it appears that differences in sampling efficiency do not affect the predictive power of both implementations, since the Stan and DSP models agree on the predictive posterior distributions, as illustrated in Figure 4.20. However, as in the case of IBM stock, they fail to capture the central peaks of the distribution of the response variable.

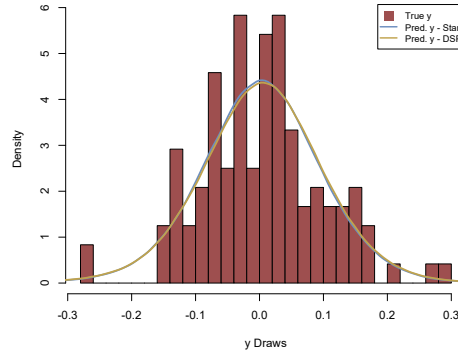


Figure 4.20: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (blue line) and Gibbs (yellow line).

MOBIL stock The MOBIL stock follows the pattern of the previous stocks, since the Stan and DSP models agree in the posterior distribution for the α and β parameters, as shown in Figure 4.21. Nevertheless, the median of the β parameter exhibits a different trend than the usual, as it does not present an increasing trend - remains constant over time after time step $t = 10$. In

addition, the β parameter exhibits an unusual behaviour at the uncertainty level; at certain time steps, such as $t = 25$, $t = 75$, and $t = 80$, the 95% HPD credible intervals spike to values much higher and lower than the posterior median. This is particularly noticeable in the DSP model and it is also possible to note that the DSP model produces a more uncertain posterior, as shown by the HPD credible intervals and Table 4.9. This could mean that the DSP model is more affected by the oil market instability, since, during the period under study, many critical events occurred in that market, for instance, the well-known 1986 oil-crash [9]. It is worth highlighting that, once again, the Stan model produces a smoother median posterior for β , it remain almost constant, while the DSP median vary.

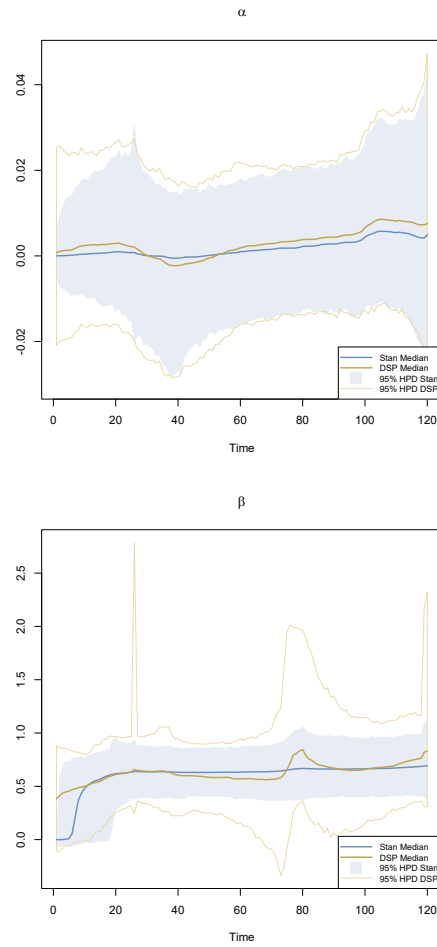


Figure 4.21: Inferred posterior over time for α_t (top) and β_t (bottom) in the CAPM model for the MOBIL stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

Parameter	Model	
	Stan	DSP
α	0.04	0.04
β	0.58	0.98

Table 4.9: Mean of the width of the HPD credible intervals (MOBIL Stock - CAPM)

Furthermore, the observation error standard deviation σ_ϵ also confirms the fact that the Stan and DSP models agree in the posterior distribution; the parameter posterior medians fluctuate around 0.06 and follow a similar trajectory, and the 95% HPD credible intervals are also similar, as shown in Figure 4.22. The abrupt jumps in the medians and credible intervals match the spikes in the credible intervals of the β parameter previously seen, however the time step $t = 25$ is the most evident. This suggests that the Stan and DSP models are sensitive to the market changes.

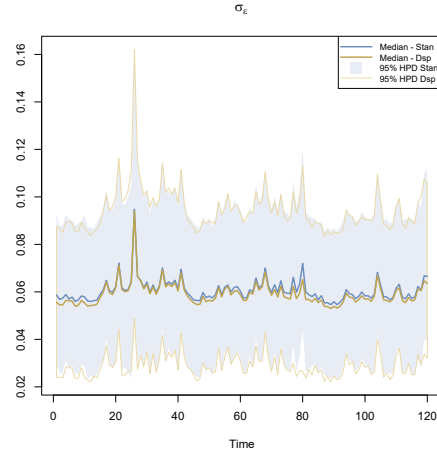


Figure 4.22: Inferred posterior over time for σ_ϵ (SV process) and 95% HPD credible intervals. The thicker lines represent the posterior medians, while the shaded area and thinner outer lines represent the credible intervals. Each model is represented by a colour, as shown in the legend.

Considering the runtime and sample efficiency, it appears that the events in the market caused some issues for the Stan model; since the average runtime until this experiment has been ten minutes, whereas for this stock it was almost four times longer, as shown in Table 4.10. The sample efficiency also indicates issues, as the N_{eff} per minute has never been so low for the Stan model, ranging approximately from 17 to 294. However, the DSP model presents the usual spread out histogram, with N_{eff} per minute ranging approximately from 38 to 3213, as illustrated in Figure 4.23. Note that even the DSP model demonstrates some issues, since the previous experiments never reported such a low minimum value for N_{eff} per minute. Therefore, this may indicate that the Stan and DSP models were affected by the complexity of this data, but in general, the Stan model appears to be the most affected.

It is worth highlighting that the Stan model was tested more than one time, since this excessive run time and the low N_{eff} per minute could be due to bad initialization or poor seeding. Nevertheless, the results were consistent across the multiple iterations. Moreover, once again, the DSP model appears to be more efficient.

Model	Stan	DSP
Time	41.30	3.13

Table 4.10: Total runtime (in minutes) (MOBIL stock - CAPM)

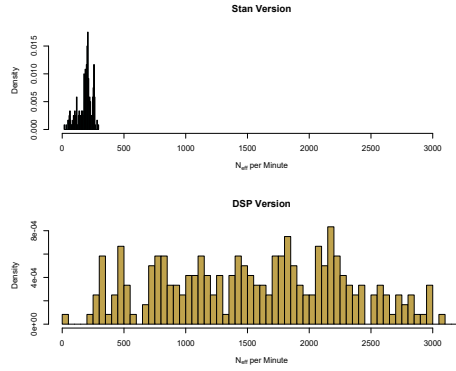


Figure 4.23: Distribution of N_{eff} per minute. The histogram for the Stan model represent the selected tuning (adapt_delta - 0.995), and follow the colour scheme presented in Table 4.1. Whereas the histogram for the DSP represents the implementation with the modified offset.

In addition, from Figures 4.24 and 4.25 it is possible to infer that the Stan and DSP models struggle around the time steps $t = 25$ and $t = 80$, for both parameters. These may indicate that the data is very challenging; therefore, the Stan and DSP models may bear less responsibility for the sampling difficulties observed. It is also important to note that the Stan version seems to recover faster from the drops and appears to be able to maintain a more stable trajectory overall.

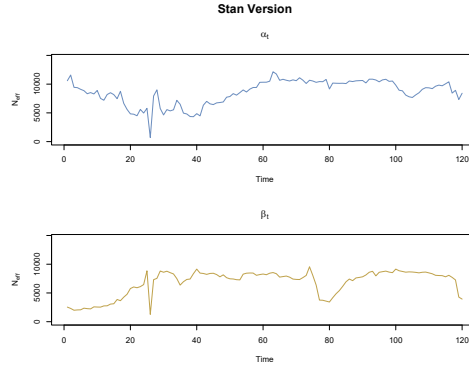


Figure 4.24: Trajectory of N_{eff} over time from the Stan model. Each parameter is represented by a colour: the blue colour represents the α_t and the yellow colour represents the β_t .

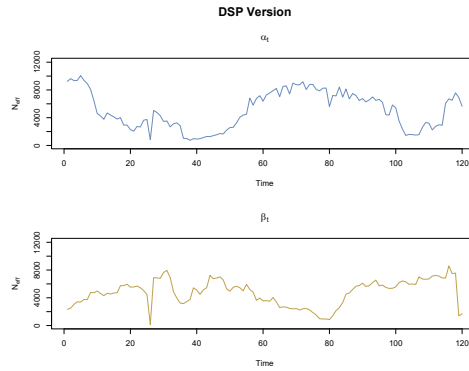


Figure 4.25: Trajectory of N_{eff} over time from the DSP model. Each parameter is represented by a colour: the blue colour represents the α_t and the yellow colour represents the β_t .

Regardless of the complexity of the data, and consequently, of issues related to sampling efficiency, the predictive power of the Stan and DSP models is still acceptable, as both predictive posteriors capture the response variable distribution, as illustrated in Figure 4.26. Compared with the previous stocks, it can be inferred that this case provides the best approximation, at least in the central region of the response distribution.

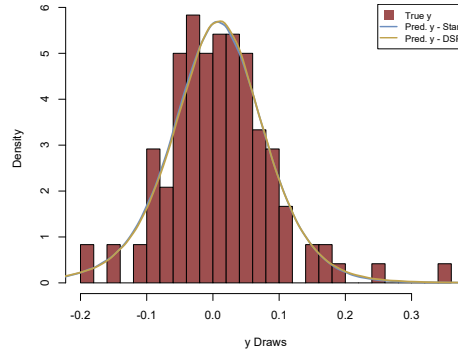


Figure 4.26: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (blue line) and Gibbs (yellow line).

Fama-French model results

IBM Stock As illustrated in Figure 4.27, the Stan and DSP models produce a similar posterior distribution for the β_1 parameter, whereas, for the α parameter, both models agree on the general trend but disagree in terms of variability; the α median from the DSP model seems to fluctuate more around zero than the one from the Stan model - the latter remains more stable. This extra variability of the median may occur due to the offset of the DSP model. In general, the DSP model presents a wider credible interval for both parameters, as indicated by Table 4.11.

Interestingly, both parameters maintain the general trend from the first real-world data experiment, nevertheless the behaviour is not exactly the same; this is expected, as this experiment uses more explanatory variables, therefore these new variables may absorb some of the variation.

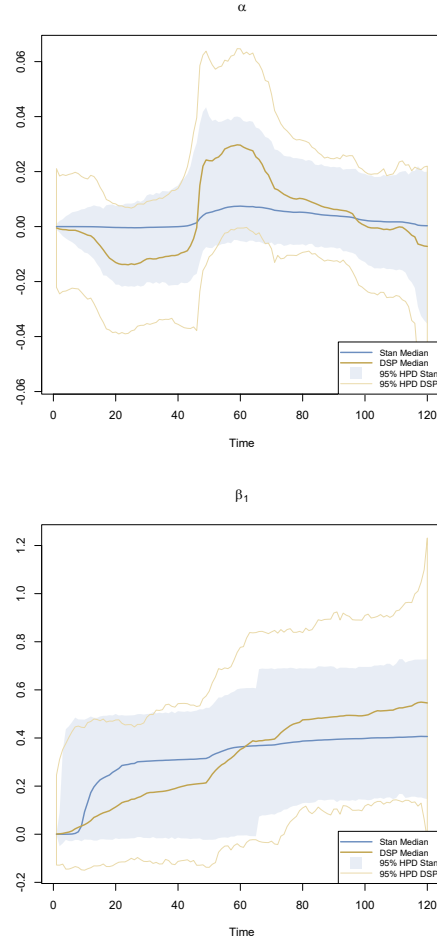


Figure 4.27: Inferred posterior over time for α_t (top) and $\beta_{1,t}$ (bottom) in the Fama-French model for the IBM stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

In the case of the new parameters, the β_2 parameter present a similar posterior distribution for the Stan and DSP models. On the other hand, with regard to β_3 , the Stan and DSP models do not agree on the posterior for the parameter, at least until $t = 20$; the median from the DSP model drops heavily to values around -0.02, while the median from the Stan model decreases only slightly (remaining around zero). Furthermore, the DSP model presents a less stable median for both parameters, as shown in Figure 4.28; once again, this may be attributed to the offset in the DSP model. Moreover, on average, the 95% HPD credible intervals of the DSP model are wider intervals (Table 4.11), as usual.

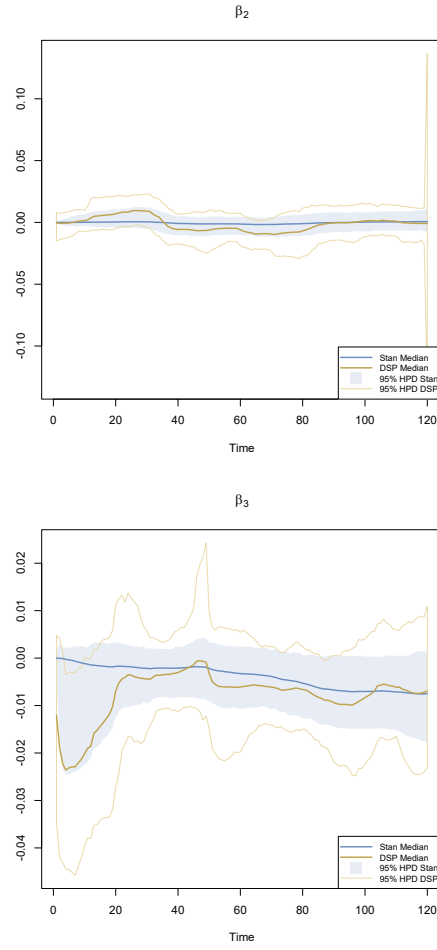


Figure 4.28: Inferred posterior over time for $\beta_{2,t}$ (top) and $\beta_{3,t}$ (bottom) in the Fama-French model for the IBM stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

Parameter	Model	
	Stan	DSP
α	0.03	0.05
β_1	0.54	0.73
β_2	0.01	0.03
β_3	0.02	0.03

Table 4.11: Mean of the width of the HPD credible intervals (IBM Stock - Fama-French model)

Considering the observation error standard deviation σ_ϵ , the Stan and DSP models appear to produce similar posterior distributions as well, as shown in Figure 4.29. The trajectories of the medians are almost identical; the main differences are in the scale, the median of Stan fluctuates around 0.05, while the median of DSP fluctuates around 0.04, and in the width of the credible intervals; on average, Stan's credible intervals have a width of 0.04, while those of DSP have a width of 0.03.

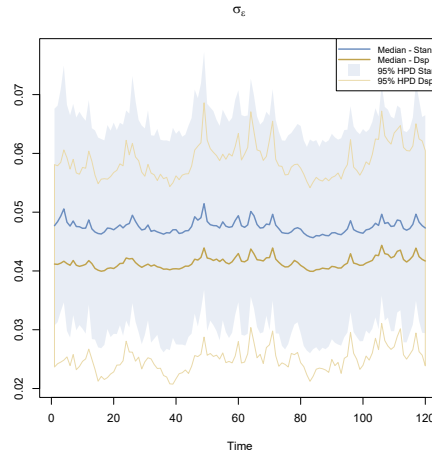


Figure 4.29: Inferred posterior over time for σ_ϵ (SV process) and 95% HPD credible intervals. The thicker lines represent the posterior medians, while the shaded area and thinner outer lines represent the credible intervals. Each model is represented by a colour, as shown in the legend.

As usual, the Stan model has a higher computational time; it takes around 37.45 minutes to run, whereas the DSP model only needs around 4.37 minutes, as specified in Table 4.12. Regarding the sampling efficiency, the Stan model presents a very narrow histogram, with values ranging from approximately 37 to 320. Whereas the DSP model presents a spread out histogram, with values ranging from approximately 137 to 4551. Thus, the DSP model appears to have a better sampling efficiency as the minimum value of N_{eff} per minute is greater than the minimum value of the Stan model, and the maximum value is much higher. However, since the Stan model concentrate the values a lot more, this may indicate that the efficiency of the model does not change depending on the parameter that is being sampled. On the other hand, the spread out histogram of the DSP model may indicate that the sampling efficiency is very variable, depending on the parameter being sampled.

Compared with the first real-world data experiment, the total run time of the Stan model increased from 13.4 minutes to 37.45 minutes, as might be expected due to the extra parameters. Surprisingly, the run time of the DSP model decreased by 41 seconds. In terms of sampling efficiency, in this experiment, the DSP model has a better sampling efficiency when compared to the Stan model; note that, in the first real-world data experiment, this result was inconclusive.

Model	Stan	DSP
Time	37.45	4.37

Table 4.12: Total runtime (in minutes) (IBM stock - Fama-French model)

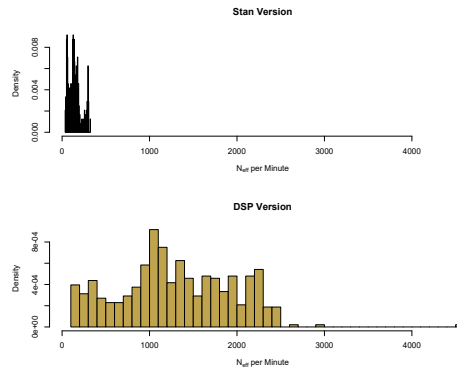


Figure 4.30: Distribution of N_{eff} per minute. The histogram for the Stan model represent the selected tuning (adapt_delta - 0.995), and follow the colour scheme presented in Table 4.1. Whereas the histogram for the DSP represents the implementation with the modified offset.

Despite the differences between the Stan and DSP models, their predictive performance appears to be similar, as shown in Figure 4.31. Along with that, the kernel density estimates of the Stan and DSP models are very similar and follow the empirical distribution of the response variable reasonably well; they only fail to capture the central spike and the right tail of the empirical distribution. These results are consistent with those of the first real-world data experiment, which may indicate that adding the variables SMB and HML did not improve the predictive performance of Stan and DSP models.

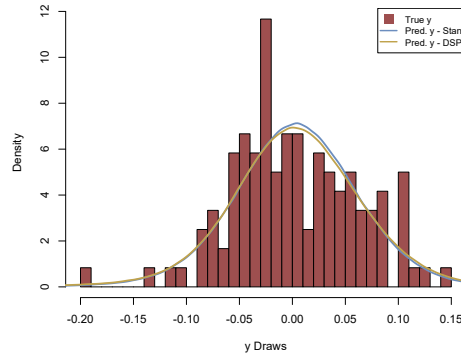


Figure 4.31: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (blue line) and Gibbs (yellow line).

DELTA Stock For the first time, the Stan and DSP models strongly disagree on the posterior of the β_1 parameter; the β_1 posterior median from the Stan model remains almost constant around zero, while the median from the DSP model shows an increasing trend. Regarding the α parameter, the usual behaviour is observed, the Stan and DSP models agree on the posterior, as illustrated by Figure 4.32. Furthermore, as usually the DSP model produced posteriors distributions with wider 95% HPD credible intervals, as presented in Table 4.13.

Apparently, the introduction of the variables SMB and HML may have absorbed the variation previously explained by the β_1 parameter; this is more clear in the Stan model, since the posterior from the DSP model presents a trend similar to that observed in the first real-world data experiment.

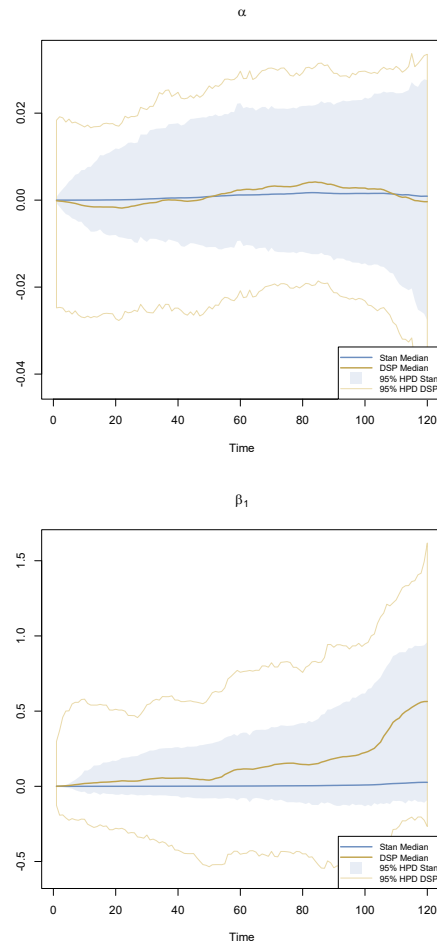


Figure 4.32: Inferred posterior over time for α_t (top) and $\beta_{1,t}$ (bottom) in the Fama-French model for the DELTA stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

In general, with regard to the new parameters, the Stan and DSP models appear to agree on both posterior distributions. Nevertheless, once again, the DSP model produces more variable posterior distributions, as indicated by the posterior medians illustrated in Figure 4.33 and Table 4.13. One example of the greater variability of the DSP model is the drop of the credible interval at $t = 60$; while Stan remains close to -0.02, the DSP reached almost -0.08.

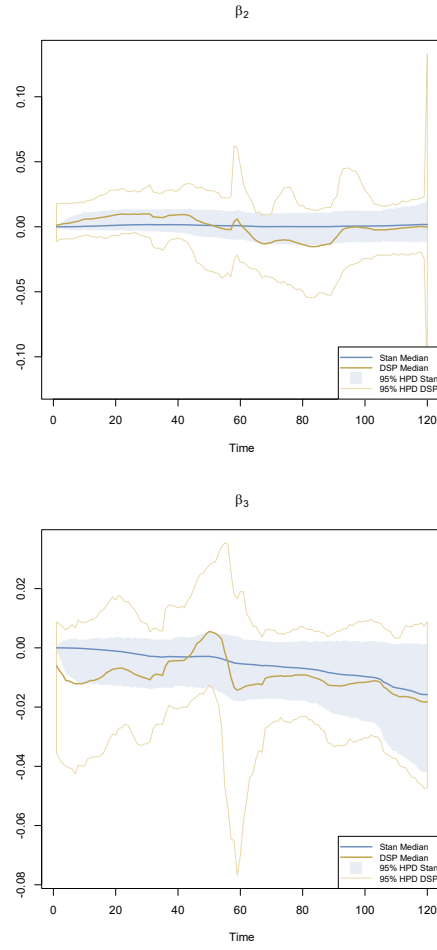


Figure 4.33: Inferred posterior over time for $\beta_{2,t}$ (top) and $\beta_{3,t}$ (bottom) in the Fama-French model for the DELTA stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

Parameter	Model	
	Stan	DSP
α	0.03	0.05
β_1	0.48	1.17
β_2	0.02	0.05
β_3	0.02	0.04

Table 4.13: Mean of the width of the HPD credible intervals (DELTA Stock - Fama-French model)

The posterior distribution of observation error standard deviation σ_ϵ produced by the Stan and DSP models seems to be in agreement. Moreover, as in IBM stock, the trajectories of the medians of the posterior distributions appears to be similar, with the only difference being the scale; the median from the Stan model fluctuates around 0.09, whilst the median from the DSP model fluctuates around 0.08, as illustrated in Figure 4.34. In terms of the 95% HPD credible interval, on average, the credible intervals of the Stan model are wider by 0.01.

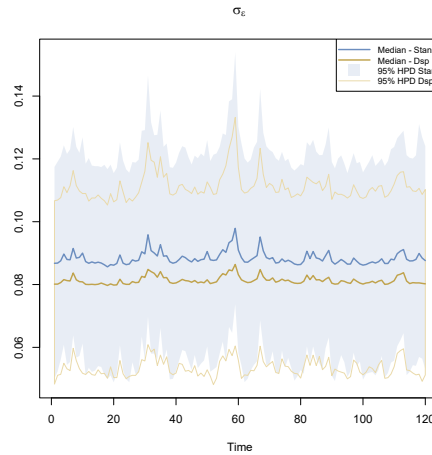


Figure 4.34: Inferred posterior over time for σ_ϵ (SV process) and 95% HPD credible intervals. The thicker lines represent the posterior medians, while the shaded area and thinner outer lines represent the credible intervals. Each model is represented by a colour, as shown in the legend.

Once again, the Stan model performed worst regarding the run time (Table 4.14) and sampling efficiency; as reported in Figure 4.35 the minimum value of N_{eff} per minute for the DSP model is higher than for the Stan model ($253.78 > 115.93$), whilst the maximum value is also higher ($11042.25 > 525.68$). Thus, the DSP sampling process is more efficient. It is worth noting that the value 11042.25 has a very low frequency, as shown in the DSP histogram; however, the maximum value of N_{eff} per minute with a reasonable frequency is still much higher than the maximum value of the Stan model ($2800 > 115.93$).

These results are consistent with those of the first real-world data experiment, as the DSP model was also faster and showed better sampling efficiency. It is important to note that the histogram from the Stan model does not present the separate blocks any more. This change may be explained by the transformation of the parameter space; by adding two variables the

values of N_{eff} per minute may become more spread out or even overlap, therefore a continuous spread out histogram is more likely.

Model	Stan	DSP
Time	23.88	4.07

Table 4.14: Total runtime (in minutes) (DELTA stock - Fama-French model)

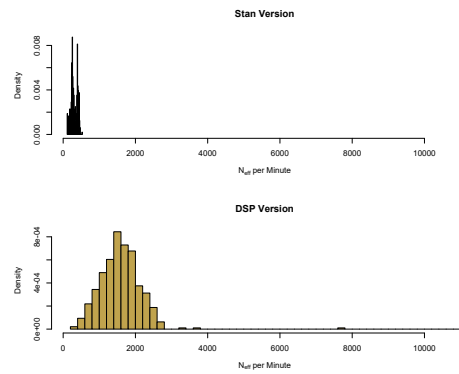


Figure 4.35: Distribution of N_{eff} per minute. The histogram for the Stan model represent the selected tuning (`adapt_delta` - 0.995), and follow the colour scheme presented in Table 4.1. Whereas the histogram for the DSP represents the implementation with the modified offset.

As with the IBM stock, the results were very consistent with those of the first real-world data experiment; the kernel density estimates are very similar in both experiments. Consequently, Figure 4.36 also exhibits that neither kernel density estimate captures the central peaks of the empirical distribution. Thus, it appears that the additional variables do not significantly improve the predictive performance.

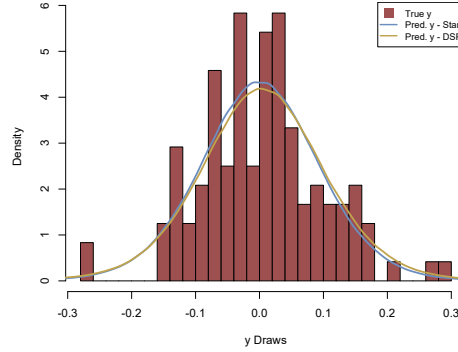


Figure 4.36: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (blue line) and Gibbs (yellow line).

MOBIL Stock The DSP and Stan models agree on the posterior distribution of the α parameter, and as usual, the median of both posteriors lies around zero. On the other hand, the posterior distribution of the β_1 parameter is not at the same level of agreement; the medians follow the same general trend: increasing trend until $t = 20$, after that time step, the Stan median stabilizes around the value 0.9, whilst the DSP median fluctuates around the same number, as illustrated in Figure 4.37. Moreover, the DSP credible intervals also exhibit similar peaks in $t = 25$ and $t = 80$, as in the first experiment with MOBIL stock; the DSP credible intervals also show the usual wider intervals, as presented in Table 4.15. It is worth highlighting that both posterior distributions closely follow the posterior distributions inferred in the first real-world data experiment. This could suggest that the introduction of the variables SMB and HML may not have absorbed the variation previously explained by the parameters α and β_1 .

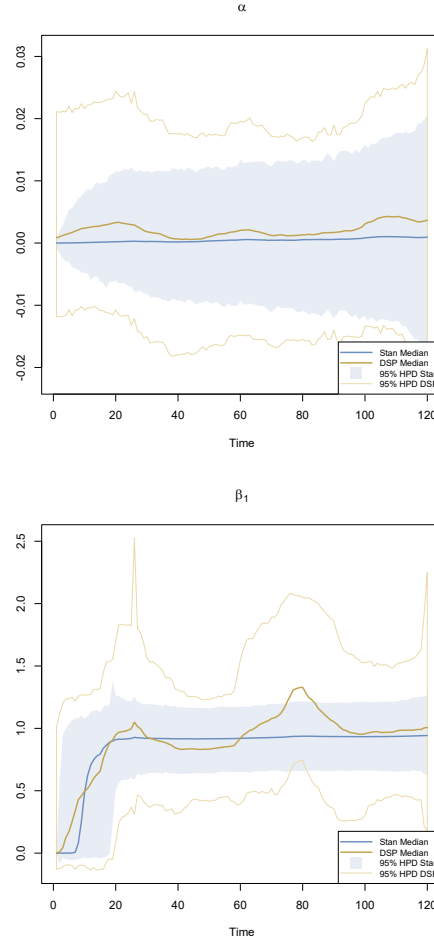


Figure 4.37: Inferred posterior over time for α_t (top) and $\beta_{1,t}$ (bottom) in the Fama-French model for the MOBIL stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

The new parameters seem to replicate the same behaviour of the parameters discussed above, as shown in Figure 4.38; more specifically, the Stan and DSP models produced a very similar posterior distribution for the β_2 parameter, whereas for β_3 , the posterior distribution is not at the same level of agreement. Nevertheless, in both posteriors, it is possible to notice the peaks and/or drops on the credible intervals of the DSP model at $t = 25$. Furthermore, the DSP model produced wider credible intervals for all parameters, as presented in Table 4.15, and less stable posterior medians.

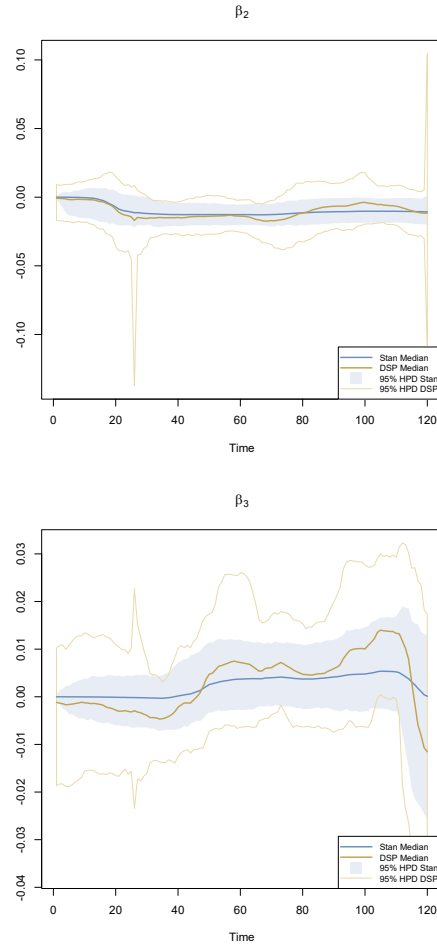


Figure 4.38: Inferred posterior over time for $\beta_{2,t}$ (top) and $\beta_{3,t}$ (bottom) in the Fama-French model for the MOBIL stock. Results from both Stan and dsp package are shown, as shown in the legend. The thicker lines represent the posterior median, while the shaded area and thinner outer lines represent the 95% HPD credible intervals.

Parameter	Model	
	Stan	DSP
α	0.02	0.04
β_1	0.63	1.25
β_2	0.02	0.04
β_3	0.01	0.03

Table 4.15: Mean of the width of the HPD credible intervals (MOBIL Stock - Fama-French model)

This may suggest that even for a more complex model (Fama-French model), the time steps mentioned before could be challenging. To better understand this phenomenon, the trajectories of N_{eff} per minute were plotted, as in the first real-world data experiment. A similar conclusion can be inferred, since the Stan and the DSP model struggle mainly at $t = 25$; in addition, for some parameters, such as β_1 and β_2 , the DSP model also struggles around $t = 80$. It is worth noting that, as in the first real-world data experiment, the Stan model exhibits a more stable trajectory and appears to recover faster from drops, as illustrated by Figures 4.39 - 4.40.

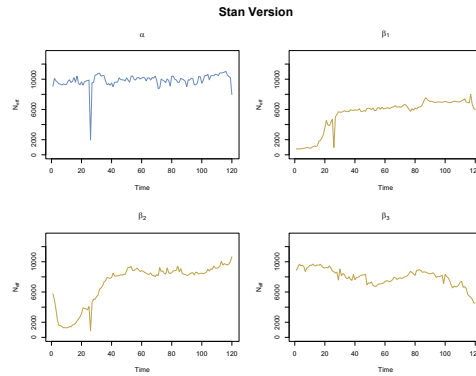


Figure 4.39: Trajectory of N_{eff} over time from the Stan model. Each parameter is represented by a colour: the blue colour represents the α_t and the yellow colour represents the $\beta_{k,t}$, $k = 1, 2, 3$.

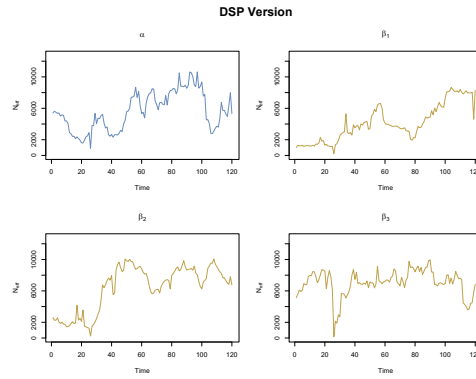


Figure 4.40: Trajectory of N_{eff} over time from the DSP model. Each parameter is represented by a colour: the blue colour represents the α_t and the yellow colour represents the $\beta_{k,t}$, $k = 1, 2, 3$.

Since the CAPM and Fama-French model struggle to fit the data in those specific time steps, this may suggest that the sampling difficulties observed

are probably due to the complexity of the data, rather than to the sampling efficiency of the DSP and Stan models.

Furthermore, the posterior distribution of the observation error standard deviation σ_ϵ also indicates agreement between the Stan and DSP models. As is customary in this section, the posterior medians differ mainly in scale; the Stan median fluctuates around 0.58, whereas the DSP median fluctuates around 0.55. In terms of the width of the credible intervals, on average, they only differ by 0.004 units. As in the previous figures, Figure 4.41 exhibits the characteristic peaks on the medians and on the credible intervals, at the usual time steps. Although the most evident peak is at time step $t = 25$.

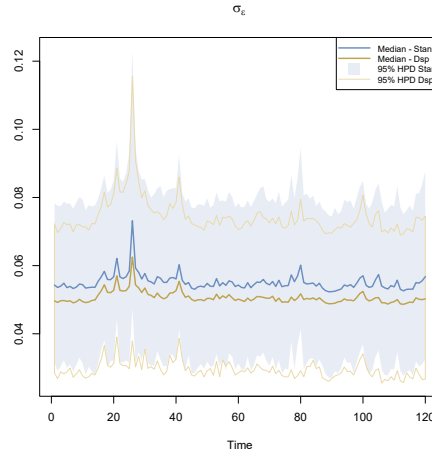


Figure 4.41: Inferred posterior over time for σ_ϵ (SV process) and 95% HPD credible intervals. The thicker lines represent the posterior medians, while the shaded area and thinner outer lines represent the credible intervals. Each model is represented by a colour, as shown in the legend.

The Stan model presented a longer run time for this stock compared to the DSP model, as specified in Table 4.16. As illustrated in Figure 4.42, the minimum N_{eff} per minute for the Stan model is lower than the minimum for the DSP model ($28.01 < 38.97$), and the maximum N_{eff} per minute for the Stan model is also lower when compared with the maximum of the DSP model ($413.13 < 2463.64$). Thus, the DSP model exhibits a better sampling efficiency.

Surprisingly, the run time of the Stan model in the second experiment was shorter than in the first real-world data experiment, while for the DSP model, the inverse happened. However, the remaining results are consistent with those from the first real-world data experiment.

Model	Stan	DSP
Time	26.73	4.32

Table 4.16: Total runtime (in minutes) (MOBIL stock - Fama-French model)

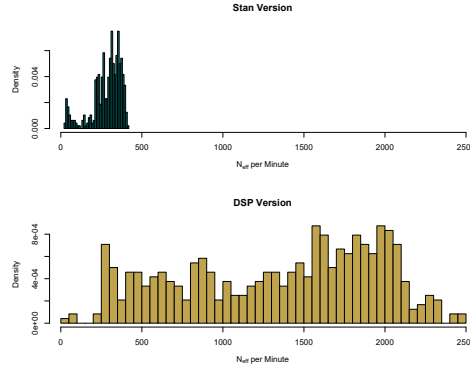


Figure 4.42: Distribution of N_{eff} per minute. The histogram for the Stan model represent the selected tuning (adapt_delta - 0.995), and follow the colour scheme presented in Table 4.1. Whereas the histogram for the DSP represents the implementation with the modified offset.

Once again, the predictive performance in the second experiment appears to be very similar to that of the first real-world data experiment. Moreover, the kernel density estimates overlap almost perfectly, as illustrated in Figure 4.43. Thus, it seems that the additional variables did not contribute to an increase in the predictive performance.

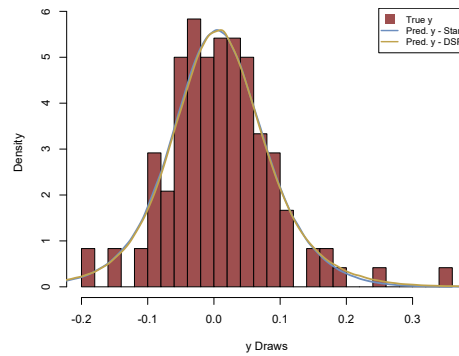


Figure 4.43: Distribution of response variable data y_t for $t = 1, \dots, T$ (histogram) and kernel density estimates of the posterior predictive draws for y draws from Stan (blue line) and Gibbs (yellow line).

5

Discussion

Stan uses the HMC algorithm to approximate posterior distributions; the main complication of the HMC algorithm is the occurrence of divergent trajectories. Current literature [23] encourages an implementation without any divergences, claiming that it is the only approach towards an optimal exploration of the posterior distribution, and consequently, increases the probability of obtaining an unbiased estimate of the posterior distribution. Nevertheless, the results obtained in this study may challenge the current literature, since the inferred posterior distributions were very similar, independently of the number of divergences. This may suggest that divergent trajectories could have a greater impact on the inferred posterior distribution when the number of algorithm iterations is low. Note that this scenario was not explored, since it falls outside the scope of the study. Moreover, increasing the value of `adapt_delta` only solved part of the divergences; it was necessary to re-parametrise the models as well, as suggested by the current literature [23]. It is worth highlighting that even when using the recommended solutions, some divergences remained.

The original study on the dynamic shrinkage process by Kowal, Matteson, and Ruppert [15] proposed an adaptive offset for the Gibbs sampling implementation. However, the results of the present study suggest that this adaptive offset may artificially increase the variability of the inferred posterior distribution; previous studies [6] have already shown that this adaptive offset has a large effect on the posterior distribution results. Thus, this study proposes a fixed offset smaller than the default offset for

the Gibbs sampling implementation, which, as previously observed, produced smoother posterior distributions with less variability.

The results obtained using synthetic and real-world data were consistent, since, in general, the Stan model produced smoother posterior distributions with lower variability than those of the DSP model. These results were to be expected, as the Stan model does not use any offset, therefore the posterior distribution is not artificially manipulated. On the other hand, the DSP model consistently presented a lower computing time and better sampling efficiency, as indicated by N_{eff} per minute. The better sampling efficiency from DSP may be unexpected, since Stan should be very efficient in high-dimensional spaces, but even in the most complex experiment (Fama-French model), the DSP model consistently outperformed the Stan model in terms of N_{eff} per minute, by a considerable margin. It is worth noting that, in the CAPM experiment, the Stan and DSP models produced a posterior distribution of the α parameter that was always around zero for every stock analysed, in line with CAPM theory [22, 16]; this indicates that Stan and DSP models can capture meaningful patterns from the data. Moreover, the differences in computing time, sampling efficiency, data complexity, and the complexity of the selected model (CAPM and Fama-French model) did not significantly affect the predictive power of the Stan and DSP models. Therefore, it is possible to implement a TVP model with a dynamic shrinkage process in Stan; depending on the objectives of the user, one of the models could be seen as the better one.

Unfortunately, there were limitations that could not be fully addressed in this study. One of these is that the proposed offset in the Gibbs sampling algorithm could not be reduced to the machine epsilon¹, as the algorithm would crash and not produce results. This is an issue because, based on the results of this study, it is very plausible to assume that, if it were possible to decrease the offset even further, this would improve the smoothness and reduce variability of the posterior distribution inferred by the DSP model. Furthermore, completely removing the divergent trajectories was not possible, despite using the suggested techniques in the current literature. Finally, the Stan software is written in a different programming language than the dsp package, and the programming language used in Stan is usually faster than the ones used in the dsp package. Therefore, the direct comparison between computation time could be seen as unfair. However, since

¹Machine epsilon is defined as 'the smallest number that may be added to 1 giving a number distinct from 1' [18].

the DSP model always outperformed the Stan model in terms of computing time; this may be considered a secondary limitation. These limitations may guide future studies, since addressing them could potentially improve the results.

A major advantage of a Stan implementation is the possibility to easily extend the model to a non-linear and/or non-Gaussian context, opening new possibilities; one possibility would be another comparison between a Stan model for count data and the model developed by Schafer and Mat-teson [21]. This model is based on the negative binomial distribution and also uses a dynamic global-local shrinkage process.

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